

Textbook priors over visual features

Can a vision–language model’s textbook knowledge stand in for labelled data?

Technical report, generated by the workflow

September 13, 2026

Contents

1	The question, and what was fixed before the numbers existed	3
2	Data and provenance	3
3	Methods as executed	4
3.1	The calls, and what was archived	4
3.2	From answers to a concept vector	4
3.3	The arms	4
3.4	The classifier, identically for C, P and C+P	5
3.5	The multi-label task	6
3.6	The permutation controls	6
3.7	The metric, the bootstrap, and n_B	6
4	Results	7
4.1	H1 — Substitution: not supported	7
4.2	H2 — The bank, not just the model: not supported	7
4.3	H3 — Scale: not supported	9
4.4	H4 — Reading the features better: not supported	9
4.5	H5 — Complement: supported	12
4.6	H6 — Effort in the frontier model: not supported	14
4.7	H7 — Capability inside the frontier family: supported	14
5	Extension: reading the arms against the published literature	14
6	Diagnostics	17
7	What this study cannot separate	17
8	Run record	19
A	The concept prompt and the zero-shot prompt, in full	19
A.1	pathmnist	19
A.2	dermamnist	23
A.3	octmnist	27
A.4	pneumoniamnist	29

A.5	bloodmnist	32
A.6	organamnist	35
A.7	breastmnist	39
A.8	retinamnist	42
A.9	tissuemnist	45
A.10	organcmnist	48
A.11	organsmnist	52
A.12	chestmnist	56
B	Sampled images, by class	59
B.1	pathmnist	59
B.2	dermamnist	61
B.3	octmnist	62
B.4	pneumoniamnist	63
B.5	bloodmnist	64
B.6	organamnist	65
B.7	breastmnist	67
B.8	retinamnist	68
B.9	tissuemnist	69
B.10	organcmnist	70
B.11	organsmnist	72
B.12	chestmnist	74
C	Sources behind the concept bank	76

1 The question, and what was fixed before the numbers existed

A multimodal language model carries textbook knowledge about what pathology looks like. This study asks what that knowledge is worth, measured in the currency a practitioner cares about: labelled images. On the 12 MedMNIST 2D benchmarks, a vision-language model scores each image against a cited bank of diagnostic visual features, and five arms are compared on one shared test sample per dataset:

- **A, zero-shot** — the model’s class distribution from the image and the class names. No labels.
- **B, textbook-only** — nearest class fingerprint from the concept bank. No labels.
- **C, concept probe** — regularised logistic regression on the concept scores, n labels.
- **P, pixel probe** — the same regression on frozen ImageNet features, the same n labels.
- **C+P, complement** — the same regression on both feature blocks concatenated, the same n labels: what the textbook adds on top of the pixels (H5).

Arms C and P differ in their features and in nothing else: the same classifier, the same regularisation search and the same labelled subsets. The five hypotheses and their decision rules were fixed in `WORKFLOW.md` before any number existed, and are reported here as they were written. Every across-dataset rule is a one-sided sign test held to one level, $p < 0.05$: an arm must win on at least 10 of the 12 datasets, or 9 of the 11 that have an arm B. The study was decided on six datasets first and extended to 12 with those six verdicts known; `WORKFLOW.md` section 2 says so, and the level reproduces the six-dataset rule exactly.

One dataset, chestmnist, is multi-label: fourteen findings that co-occur. A nearest fingerprint and a distribution over class names are not defined over such a label space, so it has arms C and P only, fitted one finding at a time, and counts for H1 alone; its AUC is the mean over findings.

One further comparison point was added after those results were in hand, and it decides nothing. The **published literature** for these same tasks is read in as a fixed ceiling, so the arms can be placed against a benchmark trained on the full training split (Section 5). It is an extension in the sense of `WORKFLOW.md` section 10: no *supported* verdict in this report reads it.

2 Data and provenance

The 12 MedMNIST v2 224-pixel 2D benchmarks (Zenodo record 10519652), read from the pinned release files whose names, checksums, split sizes and label maps are recorded in `config/medmnist.yaml` and held to the installed `medmnist` package by the smoke tier. Per dataset: a 500-image test sample shared by every arm and every model, and a 2000-image labelled pool from the official train split, both drawn with seed 0 and both capped at the official split where it is smaller — breastmnist has 156 test and 546 training images, retinamnist 400 and 1080, and the learning curve stops where their pools do.

The concept bank (`data/concepts/`) was built outside the workflow by literature review, with every feature and every class fingerprint carrying a citation. **Expert review is simulated**: no clinician has reviewed those files, and each says so. Nothing downstream should be read as if one had.

A stated limit. The test sample is random rather than stratified, so a rare class is thin: dermmnist holds six dermatofibromas and five vascular lesions of its 500. Because the AUC is the unweighted mean of one-vs-rest columns, such a column counts as much as one built on hundreds, and the absolute per-dataset AUCs should be read with that in mind. The hypotheses are paired differences on the same images, where a thin column is common mode.

3 Methods as executed

3.1 The calls, and what was archived

Each call is one chat completion carrying a system message and a user message; the user message is the rendered prompt text followed by the 224-pixel image as a lossless PNG data URI, encoded from the release array without resizing. Both prompts are pure functions of the concept bank and the pinned label map, rendered once per dataset by their own rule and hashed, so the string sent, the string hashed into every response manifest and the string printed in Appendix A are one artifact. The full text of both prompts for every dataset is in that appendix.

Calls ran at `temperature: 0` with reasoning disabled by `chat_template_kwargs.enable_thinking: false` and a budget of 512 completion tokens, in chunks of 100 images. A reply that did not parse bought exactly one retry at a doubled budget — the same prompt, the same temperature, the repair being aimed at a reply cut off mid-JSON — after which the answer was recorded as missing and never guessed. Transport failures (429 and 5xx) were retried separately with exponential backoff, up to four times, with a 120-second per-call timeout. Every attempt’s raw message is kept in the archive, including the ones that failed, so the record shows what the model said and not only what could be read from it. The archive is write-protected once written; re-querying is a deliberate act.

H4’s two readers are the exception to the reasoning setting and are described with H4 below.

3.2 From answers to a concept vector

A concept’s ordered scale maps to equally spaced values on $[0,1]$, so a level’s numeric value is its position over one less than the number of levels. This weights concepts by scale length — an adjacent miss costs 1.0 on a two-level scale and 0.25 on a five-level one — which is a consequence of the mapping and not a claim about importance, and it is why no per-concept coefficient anywhere here should be read as one. A missing answer takes the labelled pool’s median for that concept, and a binary indicator column records that it was missing, so the regression can use “the model would not say” as information rather than as a value. An indicator that would be constant carries nothing and is dropped; a concept the model never answered anywhere in the pool has no median and takes 0.5. The indicator columns are chosen on the pool and reused unchanged on the test images, because the fit and the prediction must have the same columns. Table 9 gives the resulting widths per dataset.

3.3 The arms

Arm A reads the archived zero-shot distribution straight out of the archive, one column per class in label-index order; a class the model omitted from its JSON scores zero. No softmax is applied

and none is needed: the metric ranks each class column independently.

Arm B maps each class’s fingerprint through the same level-to-number mapping, with **any** masked out, and scores a class by the negative mean absolute difference over the concepts that the fingerprint commits to *and* the image answered. A class with no overlap at all scores -1 , the worst such a mean can be: nothing speaks for it, and leaving it undefined would make the AUC undefined rather than merely uninformative. A thin fingerprint is therefore scored on few concepts, and a near-miss on any of them costs it proportionally more.

Arms C, P and C+P are one multinomial logistic regression under three feature blocks. Arm C takes the concept vectors with their missing-indicator columns; arm P takes the 512 penultimate activations of a frozen ImageNet ResNet-18 (IMAGENET1K_V1), with greyscale images repeated across three channels and only the normalisation applied — the weights’ own transform resizes to 256 and crops back to 224, which would discard the frame edge the release already provides at the right size.

Arm C+P fuses the two by concatenation, and by nothing else. The concept block (concepts plus indicators) is placed to the left of the pixel block in one design matrix, so its width is exactly the sum of the two, which Table 9 reports per dataset. That matrix is then standardised column-wise on the n labelled images — each column centred and scaled by its own mean and standard deviation over those rows, which is what puts a concept column on $[0, 1]$ and a ResNet activation on the same footing — and handed to the same regression with the same single ℓ_2 penalty over the whole matrix, the same grid, the same cross-validation and the same labelled subsets as arm P. There is no per-block weighting, no per-block penalty, no feature selection and no dimensionality reduction on either side. The point of the arm is that it differs from arm P in the presence of the concept columns and in nothing else, which is what makes the difference attributable to them; the cost of that discipline is stated with H5 below.

3.4 The classifier, identically for C, P and C+P

Features are standardised on the n labelled images and the test images are transformed by that same fit. The regularisation strength is chosen inside those same n images by stratified k -fold cross-validation, scored by mean out-of-fold accuracy, over `scikit-learn`’s C — the *inverse* penalty — on the grid $\{0.01, 0.1, 1, 10, 100\}$. There is no validation set: n labels means n labels. The fold count is the smaller of five and the smallest class count in the subset, and below two folds the grid’s median is taken without a search. The final fit uses lbfgs with a 2000-iteration cap and softmax over all classes at once. A class absent from a subset gets an all-zero score column rather than being dropped, so every arm’s score matrix has the same shape and the AUC is over the same classes; a subset containing one class only is reported as degenerate rather than given a fabricated probability.

Labelled subsets are class-stratified nested prefixes of the pool: each subset extends the one before it, so the curve is one labelling effort growing rather than unrelated draws at each size. Each is as close to the pool’s class proportions as whole numbers allow by largest remainder, with a floor of one image per class, because a subset missing a class cannot predict it. Three seeds draw three such sequences and every curve point is the mean over them.

3.5 The multi-label task

chestmnist has fourteen findings that co-occur, so arms A and B are undefined for it and are not run. Arms C, P and C+P fit the same regression one finding at a time, one-vs-rest, on the same feature blocks. Two things differ and are stated rather than left to be discovered: the subsets are stratified on whether a film carries any finding at all rather than on a class, and the penalty is chosen by out-of-fold AUC rather than accuracy, because a finding present in two percent of films makes “always negative” the most accurate classifier at every strength. A finding with no positives in a subset cannot be fitted and scores a constant column, which the metric reads as uninformative.

3.6 The permutation controls

Two, both free re-analyses of the archive. Arm B’s control attaches the same fingerprint rows to the wrong classes, so the bank still says what a textbook says and only the labels on it are shuffled; a draw equal to the identity is rejected and redrawn. Arm C’s control shuffles each concept column across images independently, so every concept’s marginal distribution survives and only its pairing with the image, and hence with the label, is destroyed. Each runs at three seeds and the reported drop is the mean over them.

3.7 The metric, the bootstrap, and n_B

The metric is `medmnist.evaluator.getAUC`: the positive column for a binary task and the un-weighted mean of the one-vs-rest columns otherwise, over classes for a multi-class task and over findings for the multi-label one. It is computed here by an equivalent average-rank formula, which the smoke tier holds to the package’s own evaluator on every task type, including on the coarse tied scores arm B produces.

One bootstrap per dataset is shared by every arm: a replicate resamples the test images once with replacement and every arm is recomputed on that same resample, so a difference between two arms is a difference of two columns of one matrix and its interval is a percentile of that. Ten thousand replicates, seed 0, 95% percentile intervals. A class with no positives in a replicate is undefined rather than counted as one half. H4’s readers are scored on a prefix of the test sample rather than all of it, so they get their own bootstrap over that prefix; two sets of images cannot share one resample.

n_B is the smallest grid point at which the pixel probe’s seed-mean AUC reaches the zero-label textbook arm. It is ordinal, not numeric: ≤ 50 records that the probe was already above at the first grid point and > 2000 that it never crossed, and the median across datasets is taken on ranks so that neither is discarded — dropping them would remove exactly the datasets carrying the most evidence.

4 Results

4.1 H1 — Substitution: not supported

Textbook features are worth a measurable number of labelled images. Supported if the median n_B is at least 100 over the 11 arm-B datasets **and** arm C beats arm P at the smallest labelled subset on at least 10 of 12 datasets. Arm C won on 2 of 12 (one-sided sign test $p = 0.9968$); the pixel probe never reached the textbook arm inside the curve on 0 datasets and started above it on 9.

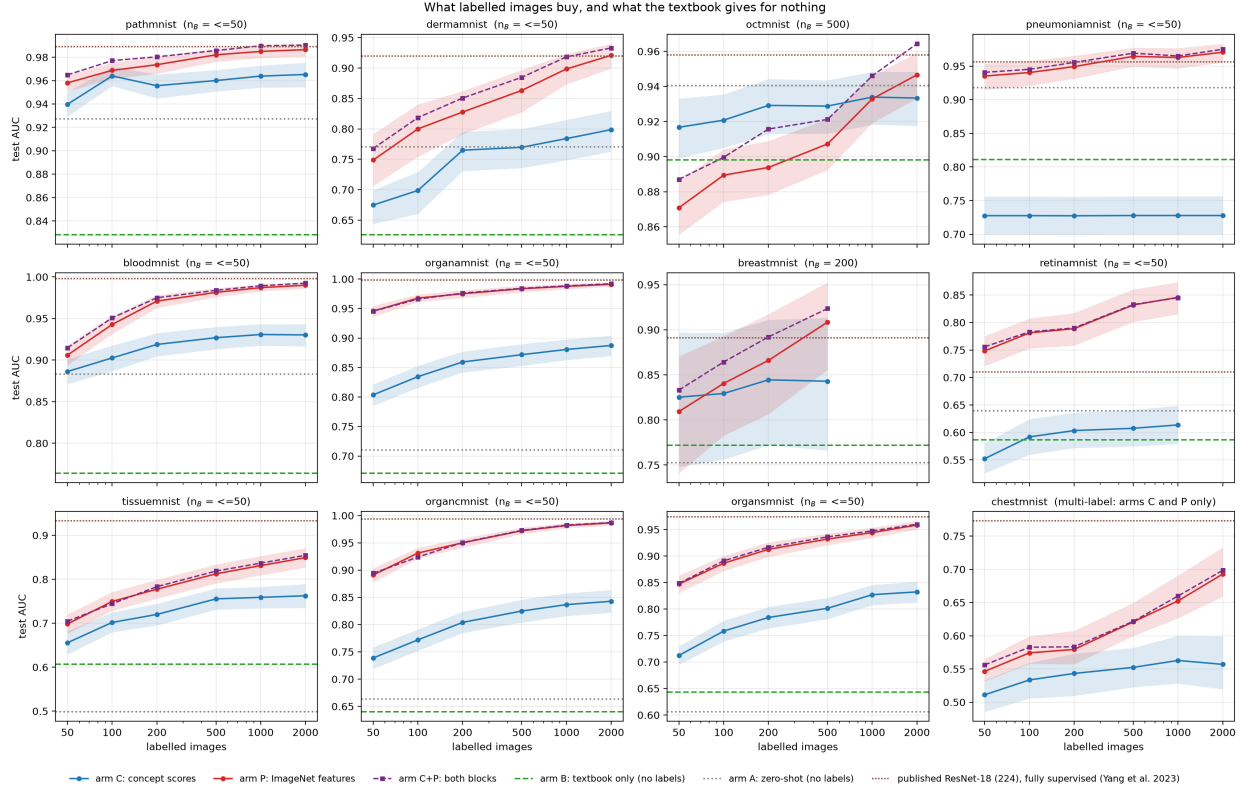


Figure 1: What labelled images buy (arms C and P, with 95% intervals; arm C+P dashed, without a band, because it tracks arm P within a few thousandths and its interval is on the paired difference in Figure 6) against what the textbook gives for nothing (arm B, dashed; arm A, dotted). Where the pixel probe crosses arm B’s line is n_B . The finely dotted brown line is not an arm: the published, fully supervised ResNet-18 (224) AUC for the same task [88], trained on the whole official training split rather than this study’s labelled pool, read here as a ceiling rather than a comparison (Section 5).

4.2 H2 — The bank, not just the model: not supported

Directing the model at cited visual features beats asking it for the diagnosis, and the bank’s structure carries the difference. Supported at 9 of 11. Arm B beat arm A on 3 of 11 datasets ($p = 0.9673$). The two prompts are separate calls on separate prompts — the concept prompt names no class and the zero-shot prompt mentions no concept, which the smoke tier checks on the rendered strings — so the comparison cannot be circular. Table 2 also reports what each arm loses when the structure

dataset	n_B	95% interval	AUC(C, n=50)	AUC(P, n=50)	difference C-P
pathmnist	≤ 50	$[\leq 50, \leq 50]$	0.940	0.958	-0.018 [-0.029, -0.007]
dermamnist	≤ 50	$[\leq 50, \leq 50]$	0.675	0.749	-0.074 [-0.134, -0.019]
octmnist	500	[100, 1000]	0.917	0.871	0.046 [0.025, 0.066]
pneumoniamnist	≤ 50	$[\leq 50, \leq 50]$	0.728	0.935	-0.208 [-0.240, -0.174]
bloodmnist	≤ 50	$[\leq 50, \leq 50]$	0.886	0.906	-0.020 [-0.038, -0.002]
organamnist	≤ 50	$[\leq 50, \leq 50]$	0.804	0.946	-0.142 [-0.159, -0.125]
breastmnist	200	$[\leq 50, > 500]$	0.825	0.809	0.017 [-0.069, 0.099]
retinamnist	≤ 50	$[\leq 50, \leq 50]$	0.552	0.748	-0.196 [-0.234, -0.158]
tissuemnist	≤ 50	$[\leq 50, \leq 50]$	0.656	0.698	-0.043 [-0.075, -0.010]
organcmnist	≤ 50	$[\leq 50, \leq 50]$	0.739	0.891	-0.152 [-0.174, -0.131]
organsmnist	≤ 50	$[\leq 50, \leq 50]$	0.712	0.847	-0.135 [-0.155, -0.114]
chestmnist	—	—	0.511	0.546	-0.037 [-0.065, -0.010]

Table 1: H1. The labelled images the pixel probe needed to reach the textbook arm, and the concept probe against the pixel probe at the smallest labelled subset. The multi-label task has no arm B and so no n_B .

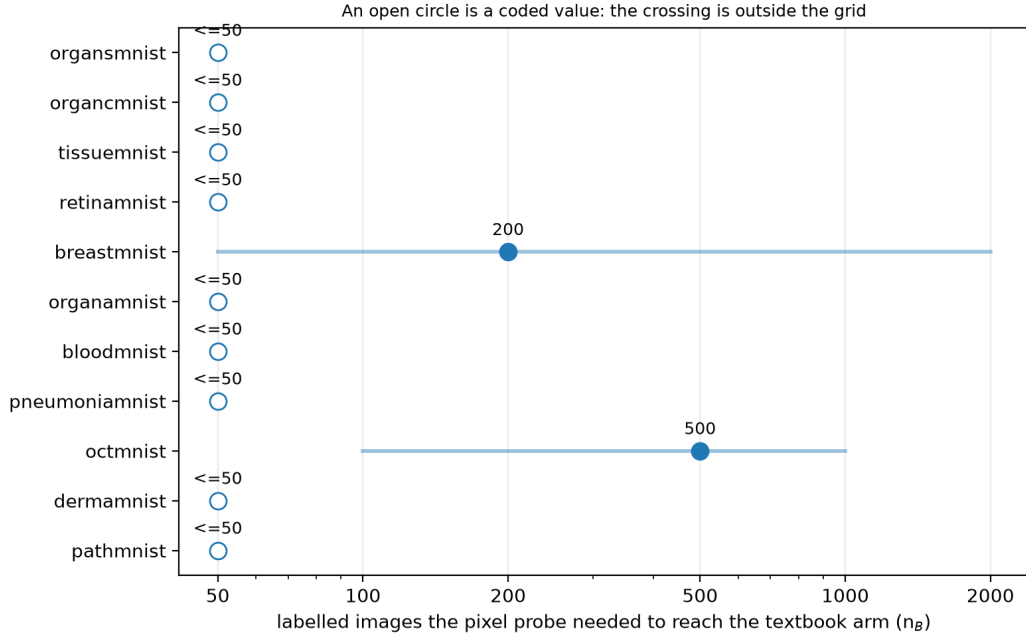


Figure 2: n_B per dataset with its bootstrap interval. An open circle marks a coded value: the crossing lies outside the grid.

it claims to read is destroyed.

dataset	AUC(B)	AUC(A)	B – A	B drop, permuted	C drop, permuted
pathmnist	0.924	0.927	-0.003 [-0.019, 0.014]	0.432 [0.415, 0.448]	0.448 [0.432, 0.463]
dermamnist	0.671	0.770	-0.099 [-0.168, -0.029]	0.209 [0.153, 0.252]	0.299 [0.255, 0.338]
octmnist	0.894	0.941	-0.046 [-0.064, -0.028]	0.519 [0.495, 0.541]	0.238 [0.219, 0.257]
pneumoniamnist	0.727	0.918	-0.191 [-0.221, -0.159]	0.454 [0.399, 0.512]	0.335 [0.294, 0.379]
bloodmnist	0.775	0.883	-0.108 [-0.136, -0.081]	0.293 [0.261, 0.321]	0.427 [0.409, 0.444]
organamnist	0.685	0.710	-0.026 [-0.059, 0.005]	0.272 [0.240, 0.306]	0.393 [0.370, 0.414]
breastmnist	0.845	0.753	0.091 [-0.001, 0.184]	0.645 [0.498, 0.771]	0.425 [0.328, 0.515]
retinamnist	0.548	0.639	-0.092 [-0.128, -0.055]	0.081 [0.047, 0.116]	0.141 [0.099, 0.183]
tissuemnist	0.642	0.499	0.143 [0.113, 0.172]	0.129 [0.097, 0.158]	0.302 [0.268, 0.334]
organcmnist	0.586	0.664	-0.077 [-0.107, -0.046]	0.174 [0.147, 0.203]	0.390 [0.364, 0.414]
organsmnist	0.617	0.606	0.011 [-0.020, 0.042]	0.175 [0.145, 0.205]	0.362 [0.336, 0.388]

Table 2: H2. The textbook arm against the zero-shot baseline, and what each arm loses when the structure it claims to read is destroyed: arm B’s fingerprints permuted across classes, arm C’s concept columns permuted across images.

4.3 H3 — Scale: not supported

The prior gets better with a bigger model. Read within family, because size and training data are confounded across families and both size steps also change quantisation. The larger model won on 7 of 11 datasets in the qwen family and 5 in the gemma family, against 9 needed in both; a Friedman test over all 4 models gives $p = 0.8964$.

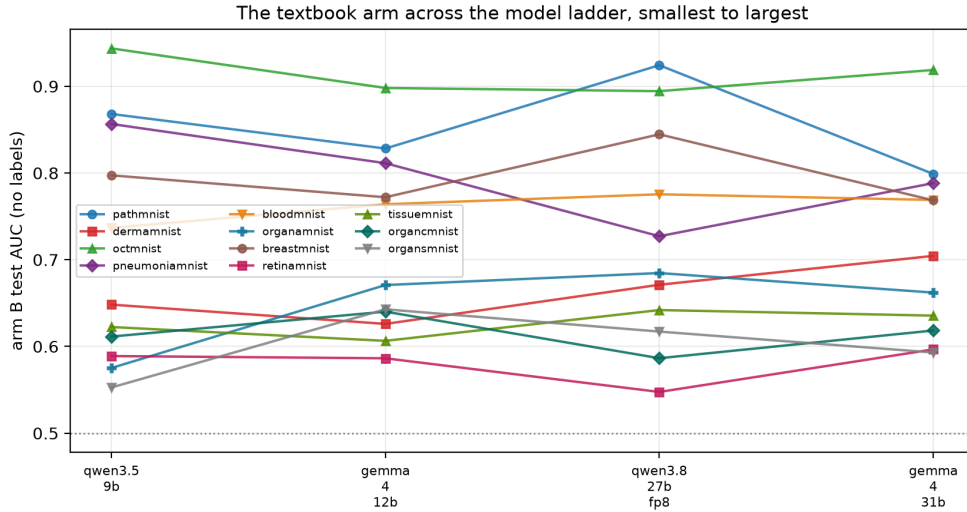


Figure 3: Arm B across the model ladder, smallest to largest.

4.4 H4 — Reading the features better: not supported

A better reader gets more class information out of the same images. Two steps, each changing exactly one thing, on the 200 images every reader was scored on: qwen3.8-27b-fp8 at no reasoning

dataset	qwen3.5-9b	gemma-4-12b	qwen3.8-27b-fp8	gemma-4-31b
pathmnist	0.868	0.828	0.924	0.799
dermamnist	0.648	0.626	0.671	0.704
octmnist	0.944	0.898	0.894	0.919
pneumoniamnist	0.856	0.811	0.727	0.788
bloodmnist	0.737	0.764	0.775	0.769
organamnist	0.575	0.671	0.685	0.662
breastmnist	0.797	0.772	0.845	0.768
retinamnist	0.589	0.586	0.548	0.597
tissuemnist	0.622	0.606	0.642	0.636
organcmnist	0.611	0.640	0.586	0.618
organsmnist	0.553	0.643	0.617	0.593

Table 3: H3. Arm B’s AUC for every model, smallest to largest. Read within family: size and training data are confounded across families, and both size steps also change quantisation.

effort, the same model asked to think at **medium (H4a)**, and gpt-5.6-terra-medium, a frontier model asked to think as hard (**H4b**). The second step cannot be run against the no-reasoning baseline, because the frontier model cannot be asked to stop reasoning; holding the effort fixed instead is what makes it a comparison of models rather than of models and effort together.

The measure is a cross-validated probe on the concept answers, not arm B and not arm C. Arm B reads the answers through the nearest-fingerprint rule, which H2 showed to be lossy, so a comparison of readers through it would be a comparison through a broken channel; arm C would need a labelled pool for every reader, which this study deliberately does not buy. The probe fits arm C’s classifier inside the scored images themselves by stratified cross-validation, with folds fixed by seed and identical across readers, and every image is scored by a fit that never saw it. It measures what the answers carry. It is not a point on the learning curve, and no n_B reads it.

The readers are the one place the call settings differ from the archive’s. Both were asked for `reasoning_effort: medium`, and where an effort is sent the thinking-off switch is not: they are two ways of saying one thing to a stack that need not agree with itself. Their token budgets are their own, sized from measured completion lengths — 2048 on the local reader, where 512 is what made thinking look impossible and a thinking answer costs about 986 completion tokens against 112 with thinking off, and 4096 on the gateway model, which is also billed at a discounted service tier. The gateway dialect takes `max_completion_tokens` rather than `max_tokens`, which is why the two readers are two rules.

Two limits on H4b. The frontier model is closed and of unknown size, so the step is capability and not parameters. And it refuses `temperature: 0` — only its served default is allowed — so it is the one reader here whose answers are sampled rather than deterministic; they would differ if re-bought, and the paired bootstrap cannot see that, because it resamples images over answers that are fixed only in the archive.

H4a won on 4 of 11 datasets and H4b on 5 of 11; both are held to the level every rule is held to, at least 9 of 11. All 9 readers are reported in Table 4 and Figure 4, not only the three the chain reads.

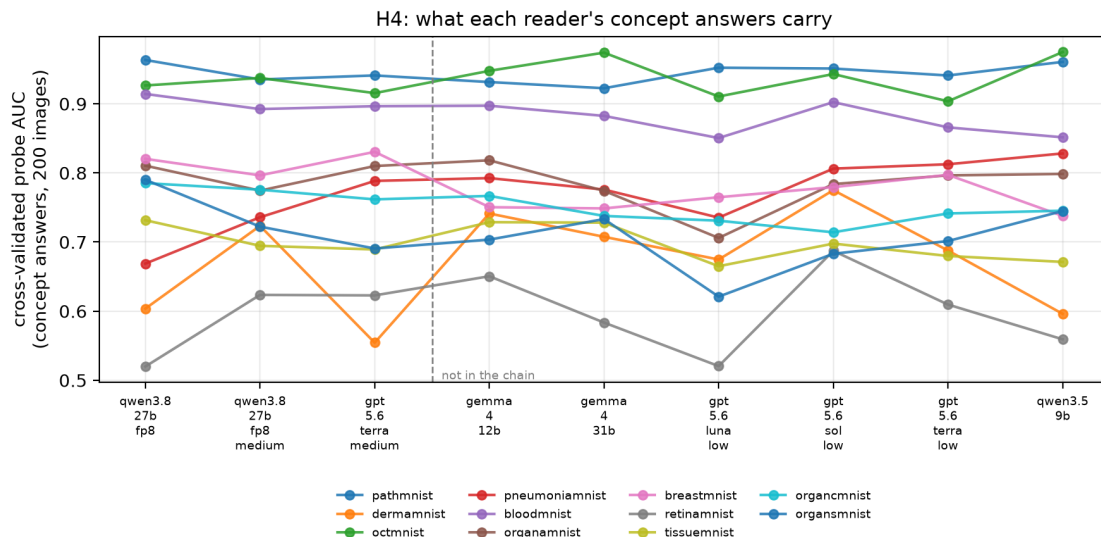


Figure 4: What each reader’s concept answers carry, by the cross-validated probe. The chain H4 is decided on sits left of the dashed rule; the remaining readers are the ladder models at no reasoning effort, shown for context.

dataset	qwen3.8-27b-fp8	qwen3.8-27b-fp8-medium	gpt-5.6-terra-medium	H4a: think – base	H4b: fro
pathmnist	0.963	0.935	0.941	-0.028 [-0.056, -0.001]	0.006 [
dermnist	0.604	0.723	0.554	0.116 [0.048, 0.177]	-0.162 [-
octmnist	0.927	0.937	0.915	0.011 [-0.015, 0.037]	-0.022 [
pneumoniamnist	0.669	0.736	0.788	0.067 [0.013, 0.124]	0.052 [
bloodmnist	0.914	0.892	0.897	-0.021 [-0.058, 0.010]	0.004 [
organamnist	0.810	0.774	0.810	-0.036 [-0.093, 0.020]	0.036 [
breastmnist	0.820	0.796	0.831	-0.023 [-0.097, 0.046]	0.033 [
retinamnist	0.520	0.624	0.623	0.104 [0.030, 0.181]	-0.001 [
tissuemnist	0.732	0.695	0.689	-0.036 [-0.091, 0.017]	-0.006 [
organcmnist	0.785	0.776	0.762	-0.010 [-0.057, 0.038]	-0.014 [
organsmnist	0.790	0.722	0.691	-0.068 [-0.118, -0.017]	-0.032 [

Table 4: H4. The cross-validated probe on each reader’s concept answers, on the 200 images every reader was scored on. H4a holds the model and adds reasoning; H4b holds the reasoning effort and changes the model. Won on 4 of 11 and 5 of 11 respectively. These AUCs are not points on the learning curve: the probe is fitted inside the same images it scores, so it measures what the answers carry rather than what a labelled model would achieve.

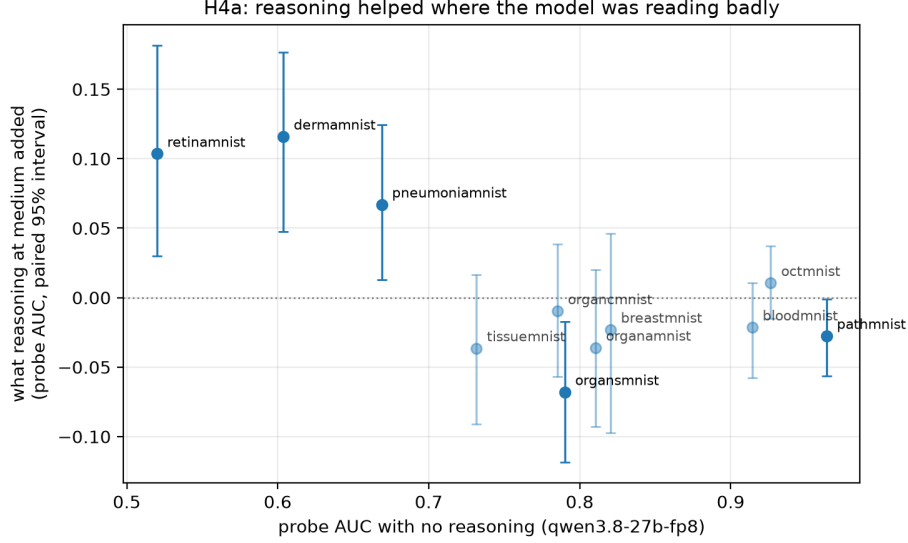


Figure 5: H4a against its baseline: what reasoning added to each dataset’s probe AUC, plotted against how well the model read the concepts with no reasoning. Solid points have intervals clear of zero. A handful of points is a pattern, not a slope, and none is fitted.

4.5 H5 — Complement: supported

The textbook adds something the pixels do not already carry. H1 asks whether the concept scores can replace the pixel features at equal labels and answers no; that is not the same as asking whether they carry anything the pixels lack. Arm C+P gives the same regularised regression both feature blocks at once: the concept block concatenated to the left of the pixel block in one design matrix, standardised column-wise on the same n labelled images and fitted with the same single ℓ_2 penalty, the same grid and the same nested subsets as arm P, with no per-block weighting and no feature selection. The fusion is described in full under Methods and its widths are in Table 9; arm C+P differs from arm P in the presence of the concept columns and in nothing else. Supported if it beats arm P at $n = 50$ on at least 10 of 12 datasets.

Arm C+P won on 11 of 12 ($p = 0.0032$), with a median difference of 0.007 AUC; 11 datasets have an interval clear of zero above and 0 below. The rule was written into `WORKFLOW.md` and `config/config.yaml` and committed before arm C+P was fitted for the first time, which is what separates it from an arm chosen after seeing what would win. It costs no calls: both feature blocks were already on disk, so the whole comparison is a re-analysis of the existing archive.

One limit, stated rather than tuned away: seven to fourteen concept columns join 512 pixel columns under a single penalty (Table 9), so the concept block is outnumbered by roughly forty to one and a real but small contribution can be shrunk away. A null here would mean no detectable gain under the classifier every other arm uses, not that the concept answers carry nothing — Table 2’s permutation controls show they carry class information. The same asymmetry cuts the other way for a positive result: a gain found under this handicap is not an artefact of giving the concept block room it was not entitled to.

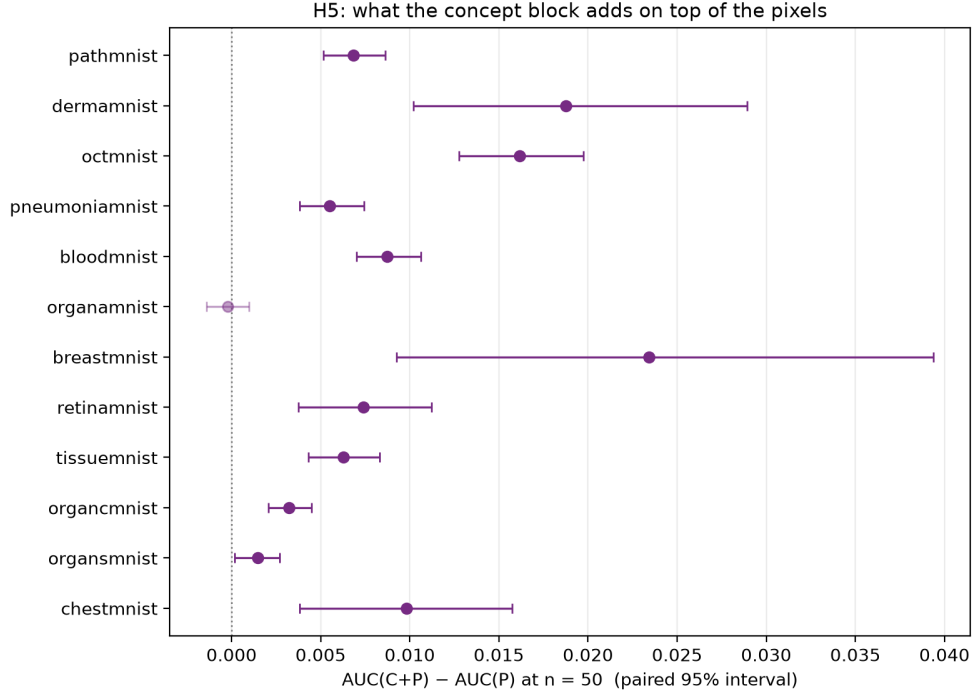


Figure 6: H5: the paired difference between the two feature blocks together and the pixel block alone, at the labelled subset the rule is decided at. Solid points have intervals clear of zero.

dataset	AUC(P, n=50)	AUC(C+P, n=50)	C+P - P at n=50	C+P - P at the largest n
pathmnist	0.958	0.965	0.007 [0.005, 0.009]	0.004 [0.002, 0.006]
dermamnist	0.749	0.768	0.019 [0.010, 0.029]	0.012 [0.005, 0.020]
octmnist	0.871	0.887	0.016 [0.013, 0.020]	0.018 [0.012, 0.024]
pneumoniamnist	0.935	0.941	0.005 [0.004, 0.007]	0.004 [0.001, 0.007]
bloodmnist	0.906	0.915	0.009 [0.007, 0.011]	0.002 [0.002, 0.004]
organamnist	0.946	0.945	-0.000 [-0.001, 0.001]	0.001 [-0.000, 0.002]
breastmnist	0.809	0.833	0.023 [0.009, 0.039]	0.015 [-0.003, 0.035] (500)
retinamnist	0.748	0.756	0.007 [0.004, 0.011]	-0.001 [-0.005, 0.003] (1000)
tissuemnist	0.698	0.705	0.006 [0.004, 0.008]	0.005 [0.001, 0.009]
organcmnist	0.891	0.894	0.003 [0.002, 0.004]	0.001 [0.000, 0.001]
organsmnist	0.847	0.849	0.001 [0.000, 0.003]	0.001 [-0.000, 0.003]
chestmnist	0.546	0.556	0.010 [0.004, 0.016]	0.005 [-0.000, 0.011]

Table 5: H5. What the concept block adds on top of the pixel block: the same classifier on arm P’s features against those features with arm C’s concatenated, standardised together under one L2 penalty. Decided at $n = 50$, where arm C+P won on 11 of 12 datasets against the 10 the rule asks for. The largest-n column is reported and decides nothing.

4.6 H6 — Effort in the frontier model: not supported

H4b’s null is a statement about the model, not about how hard it was asked to think. The same gpt-5.6-terra-medium reader, asked for **low** instead of **medium**: one setting changed, the same prompt, the same images, the same probe. **low** is the floor the gateway allows, since it rejects **minimal**, which is also why H4b could not be run against a frontier reader at no reasoning at all.

The archive motivates the question and rules out the crude version of it. Every one of the **medium** reader’s calls finished cleanly, with no truncation and no missing answer, so this is not a token-budget failure. But its reasoning tokens are 41% to 85% of its completions, and on dermamnist — where H4b loses hardest, and where a 12B open model at no reasoning reads the concepts better — it commits one concept to a single level on 93% of images where the two qwen readers sit near half.

The lower effort won on 3 of 11 datasets and the higher on 8 ($p = 0.9673$ for the lower), against 9 needed; the median difference is -0.011 AUC, with 1 intervals clear of zero for the lower effort and 0 against it. Figure 7 draws both readings of every dataset rather than only their difference, sorted by the higher effort, so the one dataset the lower effort clearly rescues can be seen to be the one the model read worst. One limit: this model is the study’s only sampled reader, so two efforts of it differ in their draw as well as their effort, and the paired bootstrap cannot see that. A second, measured at the probe before the fan-out: **low** is a small step from **medium** on this model — 109 against 132 median reasoning tokens, where the local model’s **none-to-medium** step is 112 against 986 — so this is a null about a small lever, and the gateway rejects **minimal**, so a wider one cannot be bought.

4.7 H7 — Capability inside the frontier family: supported

A more capable closed model reads cited features better. Three models of one family at **low**: gpt-5.6-luna-low, the middle rung, and gpt-5.6-sol-low. Nothing public orders these models by size, so the ladder is ordered by price — their output tokens cost 1.20, 12.00 and 20.00 dollars per million — which is the vendor’s ranking and a proxy for capability, not a parameter count. The top beat the bottom on 9 of 11 ($p = 0.0327$), median 0.052 AUC.

Figure 8 is drawn deliberately like Figure 3, the H3 ladder, because the pair is the finding: the same picture over an open-weight ladder from 9B to 31B parameters found no trend at all, and this one, ordered only by what the vendor charges, rises on most datasets. What separates these models is not what parameter count was measuring, and this study cannot say what it is — they are closed, of unknown size, training data and architecture.

5 Extension: reading the arms against the published literature

Everything in this section is a fixed comparison point, not a computation, and it decides no hypothesis. `data/literature/benchmarks.yaml` pins the AUC and ACC that [88] report for five fully supervised methods on these same 12 MedMNIST v2 tasks at the same 224-pixel resolution — trained on each dataset’s whole official training split, thousands to tens of thousands of images, not this study’s $n \leq 2000$ labelled pool. Table 7 sets every arm of this study beside the best of those five published numbers, so a reader can see how far a handful of labels and a textbook prior

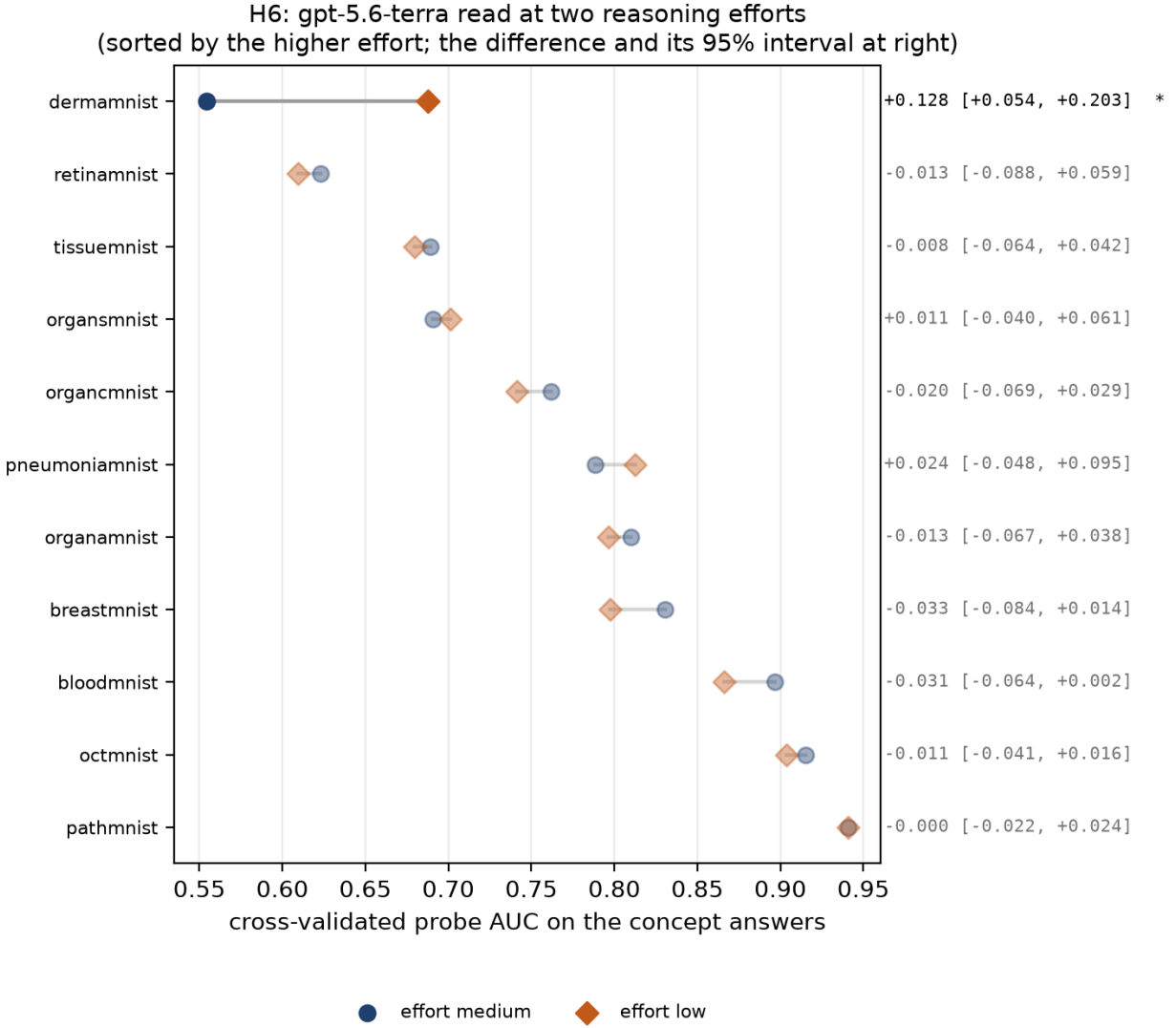


Figure 7: H6: the frontier reader at **medium** and at **low** on every dataset, joined, and sorted by the higher effort. Solid pairs have a difference interval clear of zero; the difference and its 95% interval are printed at the right. Only dermamnist, the dataset this reader read worst, is clearly rescued by lowering the effort.

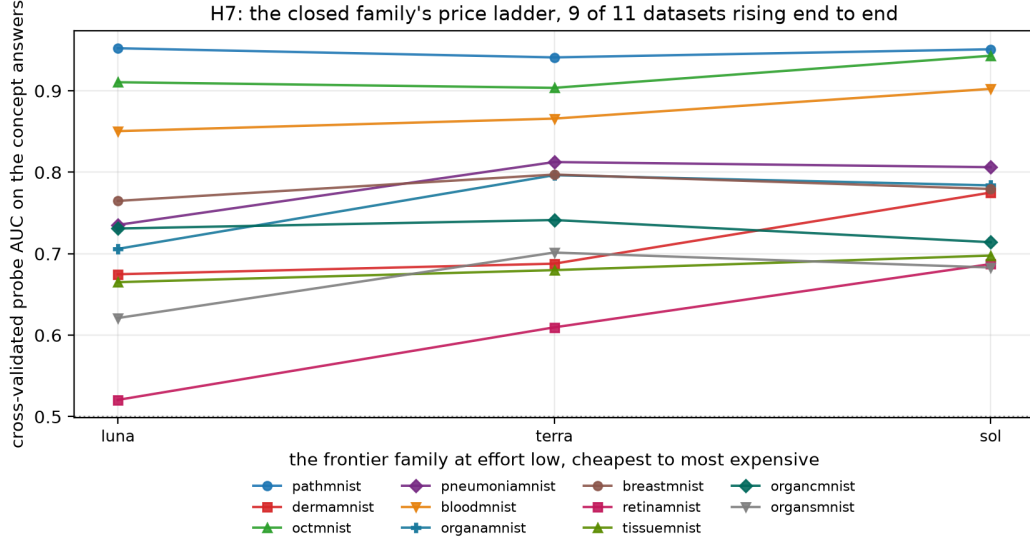


Figure 8: H7: the closed family at effort `low`, cheapest to most expensive, one line per dataset. Compare Figure 3, the same picture over open weights by parameter count.

dataset	luna-low	terra-low	sol-low	H6: terra-low – terra-medium	H7: sol-low – luna-low
pathmnist	0.952	0.941	0.951	-0.000 [-0.022, 0.024]	-0.002 [-0.026, 0.024]
dermnist	0.675	0.688	0.775	0.128 [0.054, 0.203]	0.090 [-0.014, 0.185]
octmnist	0.911	0.904	0.943	-0.011 [-0.041, 0.016]	0.032 [0.003, 0.063]
pneumoniamnist	0.735	0.813	0.806	0.024 [-0.048, 0.095]	0.071 [-0.009, 0.149]
bloodmnist	0.850	0.866	0.902	-0.031 [-0.064, 0.002]	0.052 [0.012, 0.092]
organamnist	0.706	0.796	0.784	-0.013 [-0.067, 0.038]	0.078 [0.020, 0.134]
breastmnist	0.765	0.797	0.779	-0.033 [-0.084, 0.014]	0.014 [-0.053, 0.082]
retinamnist	0.520	0.610	0.687	-0.013 [-0.088, 0.059]	0.166 [0.078, 0.253]
tissuemnist	0.665	0.680	0.698	-0.008 [-0.064, 0.042]	0.033 [-0.023, 0.088]
organcmnist	0.731	0.741	0.714	-0.020 [-0.069, 0.029]	-0.018 [-0.075, 0.040]
organsmnist	0.621	0.701	0.683	0.011 [-0.040, 0.061]	0.063 [-0.002, 0.133]

Table 6: H6 and H7, both by the cross-validated probe on the same prefix as H4. The first three columns are the frontier family at effort `low`, in price order. H6 is the same model at two efforts and won on 3 of 11 datasets for the lower effort and 8 for the higher, against 9 needed. H7 is the price ladder’s extreme step and won on 9 of 11. Price orders these closed models because nothing public orders them by size.

sit from a benchmark trained at full scale, without mistaking the gap for a finding about any arm here.

Read that way, the gap tracks the label budget and little else. The best zero-label arm on each dataset — whichever of A and B wins there — sits a median 0.115 AUC below the published best, and is within five points of it on 1 of 11 datasets. Buying 2000 labels (or a dataset’s whole pool where that is smaller) closes most of what remains: the pixel probe’s median gap is 0.009, within two points of the ceiling on 10 of 12 and at or above it on 2, while the concept probe at the same budget sits 0.129 below it. The ordering is the same one H1 reports — pixels with labels beat the textbook without them — now with an external scale attached to it.

Two cautions on reading the table. The published methods see each dataset’s whole official training split, so none of this is a same-conditions comparison, and the gaps above are not differences any bootstrap here covers: they subtract a number from another paper, with no interval. And the last column is the best of five methods per dataset, which is not always the same method; the figure instead draws the ResNet-18 (224) row alone, the one backbone trained at this study’s own input resolution, as a finely dotted line on Figure 1. Table 3 of [88] reports ACC beside AUC and `benchmarks.yaml` pins both, but only AUC appears here: this study computes no ACC for any arm, and a published ACC with nothing to set beside it would be decoration.

dataset	A	B	C (n=2000)	P (n=2000)	published	published method
pathmnist	0.927	0.924	0.965	0.986	0.989	ResNet-18 (224)
dermamnist	0.770	0.671	0.799	0.921	0.920	ResNet-18 (224)
octmnist	0.941	0.894	0.933	0.947	0.963	Google AutoML Vision
pneumoniamnist	0.918	0.727	0.728	0.971	0.991	Google AutoML Vision
bloodmnist	0.883	0.775	0.930	0.990	0.998	ResNet-18 (224)
organamnist	0.710	0.685	0.887	0.991	0.998	ResNet-18 (224)
breastmnist	0.753	0.845	0.843 (500)	0.908 (500)	0.919	Google AutoML Vision
retinamnist	0.639	0.548	0.614 (1000)	0.846 (1000)	0.750	Google AutoML Vision
tissuemnist	0.499	0.642	0.762	0.849	0.941	AutoKeras
organcmnist	0.664	0.586	0.843	0.986	0.994	ResNet-18 (224)
organsmnist	0.606	0.617	0.832	0.959	0.975	ResNet-50 (224)
chestmnist	—	—	0.557	0.693	0.778	Google AutoML Vision

Table 7: Literature reconciliation (extension, decides no hypothesis). Every arm of this study against the best of five fully supervised methods reported for the same task on the same 224-pixel release [88]. The published methods are trained on the whole official training split, thousands to tens of thousands of images, not this study’s $n \leq 2000$ pool; A and B see no labels at all, and are undefined for the multi-label task. Where a pool is smaller than the grid the subset actually reached is in parentheses. Read the last column as a ceiling for the task, not as a same-conditions comparison. ACC is pinned beside AUC in `data/literature/benchmarks.yaml` and not shown, because this study computes no ACC to set beside it.

6 Diagnostics

7 What this study cannot separate

Every source dataset is public and labelled, so “the model carries textbook knowledge” and “the model has seen this benchmark” are not distinguishable with these data. Not claimed: that the

dataset	gemma-4-12b	gemma-4-31b	qwen3.5-9b	qwen3.8-27b-fp8
pathmnist	100.0	100.0	99.8	100.0
dermamnist	100.0	100.0	100.0	100.0
octmnist	100.0	100.0	100.0	100.0
pneumoniamnist	98.2	100.0	100.0	100.0
bloodmnist	100.0	100.0	100.0	100.0
organamnist	100.0	100.0	96.4	99.4
breastmnist	100.0	100.0	100.0	100.0
retinamnist	100.0	100.0	100.0	100.0
tissuemnist	100.0	100.0	99.6	100.0
organcmnist	100.0	100.0	98.4	99.8
organsmnist	100.0	100.0	98.0	99.8
chestmnist	—	—	—	100.0

Table 8: Percentage of test images for which every concept came back with a level on that concept’s own scale. An image with any missing answer counts as incomplete; a cell more than 5% incomplete is excluded from the headline. Only the primary model scores the multi-label task.

dataset	classes	concepts	missing flags	C cols	P cols	C+P cols	labelled subsets
pathmnist	9	12	0	12	512	524	50, 100, 200, 500, 1000, 2000
dermamnist	7	12	0	12	512	524	50, 100, 200, 500, 1000, 2000
octmnist	4	9	0	9	512	521	50, 100, 200, 500, 1000, 2000
pneumoniamnist	2	8	0	8	512	520	50, 100, 200, 500, 1000, 2000
bloodmnist	8	9	0	9	512	521	50, 100, 200, 500, 1000, 2000
organamnist	11	12	2	14	512	526	50, 100, 200, 500, 1000, 2000
breastmnist	2	7	0	7	512	519	50, 100, 200, 500
retinamnist	5	10	0	10	512	522	50, 100, 200, 500, 1000
tissuemnist	8	8	0	8	512	520	50, 100, 200, 500, 1000, 2000
organcmnist	11	12	0	12	512	524	50, 100, 200, 500, 1000, 2000
organsmnist	11	12	2	14	512	526	50, 100, 200, 500, 1000, 2000
chestmnist	14	12	0	12	512	524	50, 100, 200, 500, 1000, 2000

Table 9: The design matrices as fitted. The concept block is one column per concept plus one binary missing-indicator column for each concept the primary model ever failed to answer in the labelled pool; an indicator that would be constant is dropped. The pixel block is the ResNet-18 penultimate layer. Arm C+P is the two blocks side by side, so its width is exactly their sum, and the last column is the labelled subsets that dataset’s pool could fill.

concept scores are clinically valid, that the simulated review substitutes for a clinician, or that any arm is state of the art.

8 Run record

Every result file carries a manifest: its parameters, its seeds, the git commit and dirty flag, package versions, host, thread pins and wall time. The response archive additionally records the served model name, the prompt hash and the bank hash for every call, and is write-protected once written, so re-querying is a deliberate act.

A The concept prompt and the zero-shot prompt, in full

Every call is one chat completion with a system message and a user message. The user message carries the text below followed by the image, sent as a PNG data URI (`image_url`); PNG because it is lossless, and the 224-pixel release array is encoded without resizing, so the model sees the same pixels the ImageNet encoder does. The text is a pure function of the concept bank and the pinned label map, rendered once per dataset by its own rule and hashed into every response manifest, so the string here, the string that was sent and the string the manifest names are one artifact. Levels are listed in the bank’s own order and the model is asked to copy one of them exactly; the reply is parsed against those levels and anything else is recorded as missing, never repaired into a value.

One typographic caveat: lines longer than 92 characters are hard-wrapped here to fit the page, with the original indentation carried onto the continuation. No other character is changed, and the SHA-256 printed with each prompt is of the string as sent, so the text below can be checked against `extttresults/prompts/<dataset>.json` rather than taken on trust.

The concept prompt never names a class and the zero-shot prompt never mentions a concept — H2 compares them, so their separation is checked on these rendered strings by the smoke tier rather than promised in prose. The multi-label task has no zero-shot prompt at all, because “exactly one of these categories” is false of a film carrying several findings or none.

A.1 pathmnist

Concept bank `pathmnist.yaml`, SHA-256 `5fbc404bc1c22a65....` 12 concepts, 9 classes.

Concept prompt (arms B, C and C+P). SHA-256 `905c1b6f6c1991f6c4f2e8f245397cdc...`

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of haematoxylin-and-eosin-stained colorectal histology patches, 3-channel colour, 224 x 224 pixels at about 0.5 microns per pixel. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 12 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. tissue_content

How much of the field is filled with recognisable stained tissue structure rather than plain, featureless white space?

Levels, in order, and what each looks like:

none: no stained material anywhere: flat, even white or very pale grey across the whole field

sparse: stained structure covers under about a quarter of the field; the pale rest holds no outlines

moderate: a quarter to two thirds is stained, or empty spaces fill the field but each one is walled

abundant: stained cells, fibres and nuclei fill the field, leaving no white gap wider than a nucleus

2. nuclear_density

How densely packed are the cell nuclei, seen as small dark blue-purple bodies?

Levels, in order, and what each looks like:

none: not a single dark blue-purple nuclear body anywhere in the field

few: fewer than about twenty nuclei, most lying more than several nuclear widths from the next

moderate: nuclei roughly one to three nuclear widths apart, together covering well under a quarter of the field

many: nuclei frequently touching, together covering roughly a third to a half of the field

very_many: nuclei almost confluent, covering over half the field with little visible material between them

3. nuclear_pleomorphism

Do the nuclei vary in size, shape and darkness, or are they all much alike?

Levels, in order, and what each looks like:

uniform: nuclei alike in size, shape and darkness, lying with their long axes similarly oriented

mildly_variable: largest nuclei about twice the smallest; staining slightly uneven; most still share one orientation

markedly_variable: largest nuclei three or more times the smallest, with coarse dark chromatin, prominent nucleoli and lost polarity

4. gland_formation

Are there gland-like structures, that is rings or tubes of cells enclosing an open lumen?

Levels, in order, and what each looks like:

absent: no ring or tube of cells enclosing an open lumen anywhere in the field

few: gland-like rings occupy less than half the field; the rest is solid sheets, cords or fibrous tissue

many: gland-like rings occupy more than half the field

5. gland_regularity

If gland-like structures are present, are they evenly sized, evenly spaced and smoothly outlined, or crowded, angulated and irregular in size and shape?

Levels, in order, and what each looks like:

no_glands: no ring or tube of cells encloses a lumen anywhere, so there is no gland outline to compare

regular: glands alike in size, round or oval, evenly spaced, smoothly outlined, each with an open lumen

mildly_irregular: glands differ up to about twofold in size, some branched or

- angulated, lumina still open and separate
markedly_irregular: glands back-to-back or cribriform, sharply angulated, lumina slit-like or absent, sizes differing several-fold
6. goblet_cells
Within any cell lining, are there clear rounded mucin droplets that push the nucleus to one side of the cell?
Levels, in order, and what each looks like:
absent: no clear rounded vacuole in the lining; every nucleus sits at the same level
few: clear vacuoles in under about a quarter of the lining cells, scattered singly
many: clear vacuoles in over half the lining cells, each pushing its nucleus against the cell base
7. fibre_alignment
Where elongated fibre-like structures are present, do they run in tightly packed parallel bundles, or are they wavy, loose and haphazardly oriented?
Levels, in order, and what each looks like:
no_fibres: no elongated fibre-like structure anywhere; the field holds only round cells, granular material or empty space
uniformly_parallel: almost every fibre runs on one axis, tightly packed, with no gaps or crossing bundles
mixed: two or three fibre directions present, or hyalinised bands over a fifth of the field wide crossing looser fibres
loose_and_haphazard: wavy fibres share no direction and are separated by pale amorphous ground substance
8. cytoplasm_eosinophilia
Is the material between the nuclei mostly deep pink or red, or mostly pale, grey or blue-purple?
Levels, in order, and what each looks like:
pale_or_basophilic: material between nuclei is pale grey, blue or violet, with no saturated pink anywhere
mixed: pink and pale blue-grey areas both present, neither covering more than about two thirds of the field
strongly_eosinophilic: deep saturated pink fills over two thirds of the field between nuclei, with no blue-grey zones
9. extracellular_mucin
Are there pools or wisps of pale blue-grey, structureless mucinous material lying outside cells?
Levels, in order, and what each looks like:
absent: no pale blue-grey structureless material lies outside cells anywhere in the field
focal: pale blue-grey pools cover between about a tenth and half the field
extensive: pale blue-grey structureless material covers more than half the field
10. fat_vacuoles
Does the field show large empty round or polygonal spaces bounded by thin walls, in a honeycomb arrangement?
Levels, in order, and what each looks like:
absent: no large empty rounded space; any white area is unwalled or no wider than a nucleus
few: one to three large empty rounded spaces, each many times wider than a nucleus
many: empty walled spaces in a honeycomb cover over half the field, their walls thinner

than a nucleus

11. small_round_cell_infiltrate

Are there collections of small, round, uniformly dark nuclei with almost no visible cytoplasm around them?

Levels, in order, and what each looks like:

absent: no small dark round nuclei; nuclei present are elongated, pale or vesicular

scattered: a mild, patchy increase of small dark round nuclei, singly or in small groups, not confluent

dense: small dark round nuclei form a confluent band or sheet over about a third of the field

12. necrotic_debris

Is there granular, fragmented material with broken nuclear dust and no intact cell outlines?

Levels, in order, and what each looks like:

absent: every cell has an intact outline and a whole nucleus; no granular fragmented material

focal: granular fragmented material in patches under about a third of the field, mostly inside gland lumina

extensive: granular fragmented material with scattered nuclear dust covers over a third of the field

Reply with one JSON object and nothing else: exactly 12 keys, the question names above, each value one of that question's levels copied exactly as written.

```
{
  "tissue_content": "none|sparse|moderate|abundant",
  "nuclear_density": "none|few|moderate|many|very_many",
  "nuclear_pleomorphism": "uniform|mildly_variable|markedly_variable",
  "gland_formation": "absent|few|many",
  "gland_regularity": "no_glands|regular|mildly_irregular|markedly_irregular",
  "goblet_cells": "absent|few|many",
  "fibre_alignment": "no_fibres|uniformly_parallel|mixed|loose_and_haphazard",
  "cytoplasm_eosinophilia": "pale_or_basophilic|mixed|strongly_eosinophilic",
  "extracellular_mucin": "absent|focal|extensive",
  "fat_vacuoles": "absent|few|many",
  "small_round_cell_infiltrate": "absent|scattered|dense",
  "necrotic_debris": "absent|focal|extensive"
}
```

Zero-shot prompt (arm A). SHA-256 0c72cbd01bca24940afb825ae63892a6...

You are an expert reader of medical images. You answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of haematoxylin-and-eosin-stained colorectal histology patches, 3-channel colour, 224 x 224 pixels at about 0.5 microns per pixel. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 9 categories:

1. adipose
2. background

3. debris
4. lymphocytes
5. mucus
6. smooth muscle
7. normal colon mucosa
8. cancer-associated stroma
9. colorectal adenocarcinoma epithelium

How likely is each category for this image? Give every category a probability between 0 and 1, and make the 9 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 9 keys, the category names above copied exactly as written, each value a number.

```
{
  "adipose": <probability>,
  "background": <probability>,
  "debris": <probability>,
  "lymphocytes": <probability>,
  "mucus": <probability>,
  "smooth muscle": <probability>,
  "normal colon mucosa": <probability>,
  "cancer-associated stroma": <probability>,
  "colorectal adenocarcinoma epithelium": <probability>
}
```

A.2 dermamnist

Concept bank dermamnist.yaml, SHA-256 ca0e2ccbc92eeaa2... 12 concepts, 7 classes.

Concept prompt (arms B, C and C+P). SHA-256 a4a8088519e9d45cc25fec30e1cef9f7...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 3-channel colour dermoscopic images of pigmented skin lesions. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 12 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. asymmetry_and_border

Taken together, is the lesion asymmetric about one or both of its axes, and does its pigment stop abruptly at the edge rather than fading evenly into the surrounding skin? Levels, in order, and what each looks like:

absent: The halves match across both perpendicular axes, and pigment fades gradually into the skin around the whole rim.

mild: Asymmetry across one of the two axes only, or an abrupt pigment cut-off in up to about three of the eight perimeter segments.

marked: Asymmetry across both perpendicular axes, or an abrupt cut-off over four or more of the eight segments, half the rim.

2. colour_count
How many clearly different colours are present in the lesion, counting light brown, dark brown, black, red, white and blue-grey?
Levels, in order, and what each looks like:
 one: A single one of the six hues covers the lesion; lighter and darker shading of that same hue is not a second colour.
 two: Two of the six each hold their own area, told apart where they meet rather than blending as one gradient.
 three_or_more: Three or more of the six each hold an area; white counts only where it is paler than the skin around the lesion.
3. pigment_network
Is a net-like grid of pigmented lines over a lighter background visible over any part of the lesion?
Levels, in order, and what each looks like:
 absent: No grid of pigmented lines over a lighter background; pigment is structureless, blotchy, or made of separate globules.
 present: A meshwork of brown lines enclosing lighter holes covers at least part of the lesion.
4. network_atypia
Where a net-like grid of pigmented lines is visible, are its lines thickened, darkened and broken and its holes uneven in size, rather than uniform and evenly spaced?
Levels, in order, and what each looks like:
 absent: Lines even in thickness and colour, holes near-uniform in size and thinning gradually at the rim; or no grid at all.
 present: Lines thickened, darkened and broken, holes varying widely in size and shape, the grid ending abruptly at the edge.
5. blue_white_structures
Are blue-grey or blue-white areas present in the lesion, as a hazy structureless veil, rounded grey-blue nests or globules, or fine grey dots?
Levels, in order, and what each looks like:
 absent: No blue, blue-grey or blue-white tone anywhere; the colours present are browns, tan, red or white only.
 present: A confluent blue-grey veil, rounded blue-grey nests larger than globules, or a cluster of fine grey dots.
6. arborizing_vessels
Are large, sharply focused red vessels that branch repeatedly into progressively thinner branches visible over the lesion?
Levels, in order, and what each looks like:
 absent: No branching red vessel stands out in sharp focus; any redness is diffuse or too fine to follow as a line.
 present: At least one thick, sharply focused red vessel dividing two or more times into progressively finer branches.
7. leaf_like_or_spoke_wheel_areas
Are brown-grey leaf-shaped lobules, or wheel-shaped areas with spokes radiating from a darker hub, present towards the edge of the lesion?
Levels, in order, and what each looks like:
 absent: The periphery carries no discrete bulbous brown-grey lobules and no radial spokes converging on a darker hub.

present: Discrete bulbous brown-grey extensions at the rim, or radial projections meeting at a darker central hub.

8. milia_like_cysts_and_comedo_openings

Are small round whitish-yellow spots resembling tiny cysts, or dark round plugged openings, scattered over the lesion surface?

Levels, in order, and what each looks like:

absent: The surface carries no round whitish-yellow spots and no dark round plugged openings.

few: One to about three such spots or plugs, isolated against an otherwise plain lesion surface.

many: More than about three scattered over the lesion, the whitish spots giving a starry-sky look.

9. surface_keratin_or_scale

Does the surface look keratotic - dry white-to-yellow scale, a rough brain-like (cerebriform) surface, or deep fissures separating raised ridges?

Levels, in order, and what each looks like:

absent: The surface looks smooth and glossy: no white-yellow scale, no fissures, no ridged brain-like texture.

mild: Fine white scale in patches, or a few shallow fissures between ridges, over part of the surface only.

marked: Opaque yellow-white scale, or deep fissures and ridges, over most of the surface, hiding the colour beneath.

10. red_lacunae

Are sharply outlined round or oval lagoon-like areas of red, purple, blue or black colour present within the lesion?

Levels, in order, and what each looks like:

absent: No sharply outlined round or oval lagoons; any red present is diffuse and has no defined edge.

present: One or more round-to-oval areas with a sharp border, red or purple, or blue to near-black where thrombosed.

11. central_white_patch

Is there a structureless white or scar-like patch in the centre of the lesion, with the pigmentation confined to a ring around it?

Levels, in order, and what each looks like:

absent: No structureless pale zone at the middle; colour or pigment runs across the centre of the lesion.

present: A structureless white scar-like zone at the centre, with the pigment confined to a ring outside it.

12. background_erythema

Is the background of the lesion predominantly red or pink, rather than brown, tan or the colour of surrounding skin?

Levels, in order, and what each looks like:

absent: The ground is brown, tan or the same colour as the skin around it, with no pink cast.

mild: A pink cast over part of the ground, the remainder still brown, tan or the colour of the skin around it.

marked: A red-to-pink ground over most of the lesion, clearly redder than the skin around it.

Reply with one JSON object and nothing else: exactly 12 keys, the question names above, each value one of that question's levels copied exactly as written.

```
{
  "asymmetry_and_border": "absent|mild|marked",
  "colour_count": "one|two|three_or_more",
  "pigment_network": "absent|present",
  "network_atypia": "absent|present",
  "blue_white_structures": "absent|present",
  "arborizing_vessels": "absent|present",
  "leaf_like_or_spoke_wheel_areas": "absent|present",
  "milia_like_cysts_and_comedo_openings": "absent|few|many",
  "surface_keratin_or_scale": "absent|mild|marked",
  "red_lacunae": "absent|present",
  "central_white_patch": "absent|present",
  "background_erythema": "absent|mild|marked"
}
```

Zero-shot prompt (arm A). SHA-256 6bd31912d04a0a0e42b83744fdbee12e...

You are an expert reader of medical images. You answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 3-channel colour dermatoscopic images of pigmented skin lesions. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 7 categories:

1. actinic keratoses and intraepithelial carcinoma
2. basal cell carcinoma
3. benign keratosis-like lesions
4. dermatofibroma
5. melanoma
6. melanocytic nevi
7. vascular lesions

How likely is each category for this image? Give every category a probability between 0 and 1, and make the 7 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 7 keys, the category names above copied exactly as written, each value a number.

```
{
  "actinic keratoses and intraepithelial carcinoma": <probability>,
  "basal cell carcinoma": <probability>,
  "benign keratosis-like lesions": <probability>,
  "dermatofibroma": <probability>,
  "melanoma": <probability>,
  "melanocytic nevi": <probability>,
  "vascular lesions": <probability>
}
```

A.3 octmnist

Concept bank octmnist.yaml, SHA-256 da72ca150dd4642d... 9 concepts, 4 classes.

Concept prompt (arms B, C and C+P). SHA-256 40d2f179319a81626ded17cef9776013...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 1-channel greyscale retinal optical coherence tomography (OCT) B-scans. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 9 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. retinal_layer_integrity

Are the banded retinal layers continuous and regularly stacked across the scan?

Levels, in order, and what each looks like:

intact: every band, down to the thin bright ellipsoid-zone line, runs unbroken from one edge of the frame to the other

mildly_disrupted: one or two bands blur or break over less than about a quarter of the frame width, and the stacking order stays readable throughout

markedly_disrupted: banding is lost over more than about a quarter of the frame width, or layers are so displaced that their order cannot be traced

2. foveal_contour

Does the retinal surface at the centre form a smooth shallow depression?

Levels, in order, and what each looks like:

depressed: the inner surface dips into a smooth shallow concave pit whose floor lies below the inner surface on either side

flattened: the central dip is shallow or gone, so the inner surface runs almost level across the middle of the frame

elevated: the centre of the inner surface bulges upward and stands higher than the inner surface to either side of it

3. retinal_thickness

How thick is the retina between its inner surface and the bright outer band?

Levels, in order, and what each looks like:

not_increased: the retina is at its thinnest in the centre of the frame and grows thicker towards both edges

mildly_increased: the central retina is as thick as, or a little thicker than, the retina at the edges of the same frame

markedly_increased: the central retina is more than about half again as thick as the retina at the edges of the same frame

4. intraretinal_cystoid_spaces

Are there rounded dark low-reflectivity pockets inside the retinal layers?

Levels, in order, and what each looks like:

absent: the layers between the inner surface and the bright outer band are evenly banded, with no dark rounded pocket among them

few_small: one to three dark rounded pockets, each under about a third of the retinal thickness tall, with retinal tissue both above and below it

- many_or_large: four or more such pockets, or one taller than half the retinal thickness, each still roofed and floored by retinal tissue
5. subretinal_fluid
Is there a dark empty space lifting the retina off the bright band beneath it?
Levels, in order, and what each looks like:
absent: the outermost retinal bands sit directly on the bright outer band, with no dark gap between the two
small: a shallow dark gap floored only by the bright outer band lifts the outermost retinal bands by under about a quarter of the retinal thickness
large: a dome of dark space raises the whole outer retina more than a quarter of the retinal thickness above the bright outer band
6. rpe_line_contour
Is the bright outer band smooth, gently domed in small bumps, or focally lifted?
Levels, in order, and what each looks like:
smooth: the bright outer band follows one even curve, and no bump on it is wider than about a hundredth of the frame width (63 um)
small_domed_undulations: one or more smooth rounded domes lift the band, each under about a twentieth of the frame width, the band thin and unbroken over each
large_irregular_elevation: a single lift wider than about a twentieth of the frame width, or one whose top is ragged, angular or multi-peaked rather than a smooth dome
7. subretinal_hyperreflective_material
Is there bright solid-looking material between the retina and the bright outer band?
Levels, in order, and what each looks like:
absent: the outermost retinal bands meet the bright outer band directly, with nothing bright and solid-looking interposed
small: one bright patch lies between them, under about a third of the retinal thickness tall and narrower than a twentieth of the frame width
extensive: bright material between them fills more than about a third of the retinal thickness, or runs wider than a twentieth of the frame width
8. hyperreflective_foci
Are there small bright dots or specks scattered within the darker retinal layers?
Levels, in order, and what each looks like:
absent: no discrete bright dot stands out from the darker layers lying above the bright outer band
few: up to about ten separate bright dots, each no larger than the bright outer band is thick (30 um), scattered through the darker layers
many: more than about ten such dots, often gathered into clusters or strung along one layer
9. shadowing_beneath_lesion
Do bright structures cast vertical dark columns into the tissue beneath them?
Levels, in order, and what each looks like:
absent: tissue directly beneath every bright structure is as bright as the tissue beside it at the same depth
mild: one faint dim column sits under a bright structure, no wider than that structure, and the layers it crosses can still be made out
marked: dark columns under several bright structures blank out the layers beneath them all the way to the bottom of the frame

Reply with one JSON object and nothing else: exactly 9 keys, the question names above, each

```

value one of that question's levels copied exactly as written.

{
  "retinal_layer_integrity": "intact|mildly_disrupted|markedly_disrupted",
  "foveal_contour": "depressed|flattened|elevated",
  "retinal_thickness": "not_increased|mildly_increased|markedly_increased",
  "intraretinal_cystoid_spaces": "absent|few_small|many_or_large",
  "subretinal_fluid": "absent|small|large",
  "rpe_line_contour": "smooth|small_domed_undulations|large_irregular_elevation",
  "subretinal_hyperreflective_material": "absent|small|extensive",
  "hyperreflective_foci": "absent|few|many",
  "shadowing_beneath_lesion": "absent|mild|marked"
}

```

Zero-shot prompt (arm A). SHA-256 33f95fdc2bc77927cb2557c105c2f2ce...

```

You are an expert reader of medical images. You answer with the JSON object you are asked
for and nothing else.

You are shown one image from a collection of 1-channel greyscale retinal optical coherence
tomography (OCT) B-scans. The image is 224 by 224 pixels, and comes with no clinical
context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 4 categories:
1. choroidal neovascularization
2. diabetic macular edema
3. drusen
4. normal

How likely is each category for this image? Give every category a probability between 0 and
1, and make the 4 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 4 keys, the category names above copied
exactly as written, each value a number.

{
  "choroidal neovascularization": <probability>,
  "diabetic macular edema": <probability>,
  "drusen": <probability>,
  "normal": <probability>
}

```

A.4 pneumoniamnist

Concept bank pneumoniamnist.yaml, SHA-256 316d7b97ea793b23.... 8 concepts, 2 classes.

Concept prompt (arms B, C and C+P). SHA-256 5cab4d848c7c30556a15c25e65b1889...

```

You are an expert reader of medical images. You report only what is visible in the image,
and you answer with the JSON object you are asked for and nothing else.

```

You are shown one image from a collection of 1-channel greyscale paediatric frontal chest radiographs. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 8 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. parenchymal_opacity

Is any part of either lung field whiter than the surrounding dark, air-filled lung?

Levels, in order, and what each looks like:

absent: both lung fields read uniformly dark from apex to base apart from ribs, vessels and the heart shadow

subtle: a faintly paler zone hard to separate from overlying rib or vessel shadow

definite: an area plainly whiter than the matching region of the opposite lung, with a discernible edge

2. opacity_extent

How much of the combined lung area is taken up by whiter, less dark regions?

Levels, in order, and what each looks like:

none: no zone is whiter than the surrounding lung; whiteness is confined to ribs, heart and hilar vessels

small_patch: patchy whiteness too small to fill a lobe, below the magnitude WHO requires for primary-endpoint consolidation

lobe_sized: a dense or fluffy opacity occupying a portion or the whole of one lobe

most_of_both_lungs: whiteness covering an entire lung, or spread through the greater part of both lung fields

3. opacity_density

Where the lung looks whiter, are the fine vessel lines still visible through it, or completely hidden by solid whiteness?

Levels, in order, and what each looks like:

none: vessel and airway margins stay as crisp as those in the opposite lung throughout

hazy_lines_visible: a veil of added whiteness through which vessel and airway margins can still be traced

solid_lines_hidden: homogeneous whiteness that hides vessel margins and airway walls, dense or fluffy rather than lacy

4. linear_streaky_markings

Are fine linear, streaky or net-like white lines spread through both lung fields?

Levels, in order, and what each looks like:

absent: only smoothly tapering vessel branches; no extra streaky or net-like white lines

mild: a few extra streaky lines beyond the tapering vessels, limited to one lung or to a couple of zones

marked: linear and patchy densities in a lacy net-like pattern involving both lungs

5. air_bronchogram

Inside a whiter region, are branching dark tube-like lines visible against the white?

Levels, in order, and what each looks like:

absent: no dark branching tubes stand out within any whiter region of lung

present: one or more dark air-filled bronchi traced against surrounding white airless lung

6. peribronchial_thickening

Around the central root of each lung, do the branching markings look thickened, cuffed or ring-shaped rather than fine and tapering?

Levels, in order, and what each looks like:

absent: bronchial walls beside the hilar vessels are hairline thin and end-on bronchi show no ring

mild: a few hilar branches carry a visible cuff or ring wall, the rest still fine and tapering

marked: coarse cuffed walls and ring shadows on most branches radiating from both hila

7. border_obscuration

Is the outline of the lower heart edge, or of either dome of the diaphragm, blurred or lost where lung meets it?

Levels, in order, and what each looks like:

absent: the lower heart edge and both diaphragm domes are traced as sharp unbroken lines

partial: part of the lower cardiac edge or of one diaphragm dome is blurred while the rest stays traceable

complete: the lower cardiac edge or a whole diaphragm dome is not depicted where opaque lung abuts it

8. costophrenic_angle_blunting

Is either lower outer angle between the diaphragm and the chest wall filled in or rounded rather than a sharp point?

Levels, in order, and what each looks like:

absent: both outer angles between diaphragm and chest wall taper to a sharp point

mild: one angle is rounded or shallowly filled so that its apex is no longer a point

marked: fluid fills the angle and tracks up the lateral chest wall well above the diaphragm

Reply with one JSON object and nothing else: exactly 8 keys, the question names above, each value one of that question's levels copied exactly as written.

```
{
  "parenchymal_opacity": "absent|subtle|definite",
  "opacity_extent": "none|small_patch|lobe_sized|most_of_both_lungs",
  "opacity_density": "none|hazy_lines_visible|solid_lines_hidden",
  "linear_streaky_markings": "absent|mild|marked",
  "air_bronchogram": "absent|present",
  "peribronchial_thickening": "absent|mild|marked",
  "border_obscuration": "absent|partial|complete",
  "costophrenic_angle_blunting": "absent|mild|marked"
}
```

Zero-shot prompt (arm A). SHA-256 d3285ebf39882934e707c19513fa72d2...

You are an expert reader of medical images. You answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 1-channel greyscale paediatric frontal chest radiographs. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 2 categories:

1. normal

2. pneumonia

How likely is each category for this image? Give every category a probability between 0 and 1, and make the 2 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 2 keys, the category names above copied exactly as written, each value a number.

```
{
  "normal": <probability>,
  "pneumonia": <probability>
}
```

A.5 bloodmnist

Concept bank bloodmnist.yaml, SHA-256 3aa06e349610a1b7.... 9 concepts, 8 classes.

Concept prompt (arms B, C and C+P). SHA-256 4a3f77c8ad67e6bb5d838514c83bd0ff...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of three-channel colour microscopy images of individual normal peripheral-blood cells, May-Grunwald-Giemsa stained, captured on a CellaVision DM96 analyser. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 9 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. nucleus_present

Does the cell in the centre of the image contain a nucleus, that is a distinct darker internal body, or is the whole cell body uniform with no nucleus at all?

Levels, in order, and what each looks like:

absent: the cell body is one even mass throughout, with no darker internal patch standing out from the rest

present: a distinctly darker, sharply bounded internal body sits inside the cell and contrasts with the paler cytoplasm

2. cell_size

How large is the central cell compared with the surrounding red blood cells: clearly smaller than one red cell, about the same size, or clearly larger?

Levels, in order, and what each looks like:

much_smaller: the cell spans about a quarter of a neighbouring red cell, near 2 um across

similar: the cell matches a nearby red cell in width to within about a fifth of that width

larger: the cell is plainly wider than every red cell around it, up to about twice a red cell across

3. nuclear_lobation

What is the shape of the nucleus: no nucleus at all, a single round or oval mass, a

single indented, kidney-shaped or band-like mass, two lobes joined by a strand, or three or more lobes joined by thin strands?

Levels, in order, and what each looks like:

none: the cell carries no internal outline at all, so neither a lobe nor an indentation can be sought

round: a single unbroken circular or oval mass with a smooth convex edge and no notch in it

indented: one mass whose edge is notched, kidney- or band-shaped, the notch cutting less than half way across

bilobed: two rounded segments of similar size joined by a single narrow thread of chromatin

multilobed: three to five rounded segments strung in a chain, each joined to the next by a thin thread

4. chromatin_density

How dense is the nuclear material: there is no nuclear material at all, open and finely textured with pale areas, moderately clumped, densely clumped but with a visible coarse pattern of light and dark, or so uniformly dark and structureless that no internal pattern can be seen at all?

Levels, in order, and what each looks like:

no_nucleus: the cell holds no nuclear material anywhere, so there is no interior texture to judge for density

open_fine: the nuclear interior is mostly pale with an even fine stipple and no dark clump larger than a dot

moderately_clumped: dark clumps cover under about half the nuclear area and the pale gaps between them stay plainly visible

dense_clumped: coarse violet clumps cover most of the nucleus, leaving only thin pale seams between them

pyknotic_structureless: the nucleus reads as one uniformly dark blot in which no lighter area or pattern can be made out

5. nucleus_to_cytoplasm_ratio

About what fraction of the cell area is nucleus: none of it, because there is no nucleus, less than a third (low), between a third and two thirds (moderate), or more than two thirds (high)?

Levels, in order, and what each looks like:

no_nucleus: no nuclear area is present to measure, so the cell body reads as cytoplasm from edge to edge

low: cytoplasm forms a broad rim and the nucleus covers less than a third of the cell area

moderate: the nucleus covers between a third and two thirds of the cell, with a clear rim of cytoplasm all round

high: the nucleus fills more than two thirds of the cell and cytoplasm survives only as a thin crescent

6. cytoplasm_basophilia

What colour is the cytoplasm around the nucleus: pink or almost colourless, pale grey-blue, plainly blue, or deep intense blue?

Levels, in order, and what each looks like:

pink_or_colourless: the cytoplasm is pink or almost clear, no bluer than the red cells lying around the cell

pale_grey_blue: a faint grey-blue wash, bluer than the red cells but much paler than the nucleus

blue: an even mid-blue read at a glance, clearly stronger than a grey wash but lighter

than the nucleus
deep_blue: an intense navy blue nearly as dark as the nucleus, the strongest blue
anywhere in the frame

7. granule_abundance

How prominent are granules, that is small discrete dots, in the cytoplasm: none visible,
fine and sparse or dust-like, or coarse and abundant so that they fill the cytoplasm?
Levels, in order, and what each looks like:

absent: the cytoplasm is smooth and even, showing no discrete dots against its own
colour

fine_sparse: dots are dust-like and not separately countable, so the cytoplasm looks
faintly textured rather than dotted

coarse_abundant: dots are separately countable at the cell edge and pack the cytoplasm
so that little clear ground shows

8. granule_colour

If granules are visible, what colour are they, from palest to darkest: none visible, pale
lilac or neutral, orange-red, or dark purple to black?

Levels, in order, and what each looks like:

none_visible: no dots of any colour stand out from the even tint of the cytoplasm

pale_neutral: dots are pale lilac or grey, barely different in hue from the cytoplasm
carrying them

orange_red: dots are plainly orange to red, warmer than the cytoplasm and than the
violet of the nucleus

dark_purple: dots are dark purple to nearly black, darker than the nucleus they
frequently lie across

9. granules_obscure_nucleus

Do the cytoplasmic granules lie over the nucleus and hide it: not at all, partly, or so
completely that the nuclear outline cannot be traced?

Levels, in order, and what each looks like:

absent: every part of the nuclear outline can be followed; the dots lie beside the
nucleus, not across it

partial: dots cross the nucleus and blur one stretch of its edge, while the remainder
stays traceable

marked: dots cover the nucleus so densely that no stretch of its outline can be
followed at all

Reply with one JSON object and nothing else: exactly 9 keys, the question names above, each
value one of that question's levels copied exactly as written.

```
{
  "nucleus_present": "absent|present",
  "cell_size": "much_smaller|similar|larger",
  "nuclear_lobation": "none|round|indented|bilobed|multilobed",
  "chromatin_density":
    "no_nucleus|open_fine|moderately_clumped|dense_clumped|pyknotic_structureless",
  "nucleus_to_cytoplasm_ratio": "no_nucleus|low|moderate|high",
  "cytoplasm_basophilia": "pink_or_colourless|pale_grey_blue|blue|deep_blue",
  "granule_abundance": "absent|fine_sparse|coarse_abundant",
  "granule_colour": "none_visible|pale_neutral|orange_red|dark_purple",
  "granules_obscure_nucleus": "absent|partial|marked"
}
```

Zero-shot prompt (arm A). SHA-256 b1cb28f9745d2dac6cc5d9f79e62e229...

You are an expert reader of medical images. You answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of three-channel colour microscopy images of individual normal peripheral-blood cells, May-Grunwald-Giemsa stained, captured on a CellaVision DM96 analyser. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 8 categories:

1. basophil
2. eosinophil
3. erythroblast
4. immature granulocytes(myelocytes, metamyelocytes and promyelocytes)
5. lymphocyte
6. monocyte
7. neutrophil
8. platelet

How likely is each category for this image? Give every category a probability between 0 and 1, and make the 8 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 8 keys, the category names above copied exactly as written, each value a number.

```
{
  "basophil": <probability>,
  "eosinophil": <probability>,
  "erythroblast": <probability>,
  "immature granulocytes(myelocytes, metamyelocytes and promyelocytes)": <probability>,
  "lymphocyte": <probability>,
  "monocyte": <probability>,
  "neutrophil": <probability>,
  "platelet": <probability>
}
```

A.6 organamnist

Concept bank organamnist.yaml, SHA-256 e0947840f660b66d... 12 concepts, 11 classes.

Concept prompt (arms B, C and C+P). SHA-256 9c33f158d4ae40dcf11b00af6653d4dc...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 1-channel greyscale abdominal CT crops at a fixed abdominal window; each image is a 2D bounding-box crop around one organ taken from a CT volume in the axial plane. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 12 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. body_side

Judging from the surrounding structures caught in the crop (spine, body wall, midline vessels), does the main structure lie on the patient's right, near the midline, or on the patient's left? Remember that in a cross-sectional image the patient's left is shown on the right of the picture.

Levels, in order, and what each looks like:

patient_right: the spine or the aorta sits toward the image-right of the crop, with the structure lying to the left of it on screen

midline: body wall or paired muscles lie about equally far from the structure on both sides of the image, with no side landmark nearer than the other

patient_left: the spine or the aorta sits toward the image-left of the crop, with the structure lying to the right of it on screen

2. craniocaudal_level

From the surroundings visible in the crop, at what level of the trunk was this section taken: the pelvis, the lower abdomen, the upper abdomen, or the chest?

Levels, in order, and what each looks like:

pelvic: an iliac wing, pubic bone or rounded femoral head borders the crop, and no rib is present

lower_abdominal: gas- or contrast-filled bowel loops and psoas muscle border the crop, with neither rib nor pelvic bone in view

upper_abdominal: the lowest ribs border the crop alongside solid parenchymal organs, with no pelvic bone in view

thoracic: ribs surround the crop and nearly black air-filled lung borders the structure

3. anteroposterior_position

Do the surroundings place the main structure toward the back of the body (against the spine and the back muscles), in the middle, or toward the front body wall?

Levels, in order, and what each looks like:

posterior: a vertebral body and the thick paraspinal back muscles lie immediately against the structure

central: neither the spine nor the front body wall touches the structure; both lie beyond it at the crop edge

anterior: the front body wall, with its strap-like rectus muscle or the sternum, lies immediately against the structure

4. relative_size

Compared with the landmarks caught at the edge of the crop (a vertebral body, a rib, the thickness of the body wall), is the main structure small, medium or large?

Levels, in order, and what each looks like:

small: a single landmark at the crop edge - one vertebra or one rib - appears magnified and fills much of that edge

medium: two or three separate landmarks fit along the crop edge, each still plainly resolved

large: many small landmarks crowd the crop edge at once - several ribs, vertebrae or bowel loops together

5. elongation

Is the outline of the main structure roughly circular, moderately oval, or markedly elongated and strip-like?

Levels, in order, and what each looks like:

round: the two widths of the outline differ by less than about a quarter, and the shape fills most of the frame

oval: the long width is roughly one and a half to three times the short one, leaving the frame corners empty
elongated: the long axis runs more than about three times the short one, a strip that leaves most of the frame as background

6. contour_lobulation

Is the outline smooth and even, mildly lobulated or wavy, or markedly irregular and feathery?

Levels, in order, and what each looks like:

smooth: the border runs as an unbroken curve, with no bump deeper than the thickness of the border line itself

mildly_lobulated: one or two shallow bulges or gentle waves interrupt an otherwise continuous curve

markedly_irregular: many bumps and clefts break the border, or it dissolves into a feathery fringe against the surrounding fat

7. hilar_notch

Is there a single distinct indentation, notch or wedge cut into one side of the outline where vessels appear to enter?

Levels, in order, and what each looks like:

absent: the border runs continuously round the structure with no indentation on any side

subtle: one side is flattened or dimpled, but the dent is shallower than about a tenth of the structure's width

prominent: a wedge deeper than about a tenth of the width is cut into one side, with vessels or fat running into it

8. relative_attenuation

Overall, how bright is the main structure compared with the ordinary soft tissue and muscle visible around it?

Levels, in order, and what each looks like:

much_darker: nearly black at this window, clearly the darkest tissue in the crop and far below the muscle at the frame edge

slightly_darker: a shade below the muscle at the frame edge, but plainly mid-grey rather than black

similar: no step in grey level against the muscle at the frame edge; only the outline separates the two

brighter: a clear step above the muscle at the frame edge, yet still well short of the white of bone

much_brighter: among the brightest things in the crop, approaching the white of cortical bone or of dense contrast

9. interior_homogeneity

Is the inside of the structure uniform in brightness, or does it contain clearly different regions, internal branching lines or pockets?

Levels, in order, and what each looks like:

homogeneous: one even grey fills the interior, broken by nothing more than a few thin vessel lines

mildly_heterogeneous: faint mottling or a few branching lines cross the interior, but no region differs sharply from the rest

markedly_heterogeneous: two or more distinct zones, one clearly brighter than the other, meet at sharp internal borders

10. bright_cortical_rim

Is the structure outlined by a thin rim of the very brightest, calcium-density material, brighter than any soft tissue in the crop?

Levels, in order, and what each looks like:

absent: no calcium-white line follows the border anywhere; the edge is grey meeting grey

partial: a calcium-white line brighter than any soft tissue follows part of the border and then breaks off

complete: an unbroken calcium-white line rings the whole border, the brightest thing in the crop

11. boundary_definition

How clearly is the edge of the main structure separated from what surrounds it: barely distinguishable, moderately defined, or crisp along most of its circumference?

Levels, in order, and what each looks like:

indistinct: the edge merges into the surrounding tissue over most of its length; you could not trace it confidently

moderate: the edge is traceable over roughly half the circumference and blurred or interrupted over the rest

sharp: a crisp step in brightness follows almost the whole circumference, often with a dark fat line beside it

12. adjacent_bone

Are very bright, calcium-density structures such as vertebrae, ribs or pelvic bone visible at the edge of the crop, and if so how much of the edge do they occupy?

Levels, in order, and what each looks like:

absent: nothing of calcium-white density appears anywhere along the crop edge

small: one small patch of calcium-white touches the crop edge and occupies a minor part of it

prominent: calcium-white runs along a long stretch of the crop edge, or several bony pieces border the crop at once

Reply with one JSON object and nothing else: exactly 12 keys, the question names above, each value one of that question's levels copied exactly as written.

```
{
  "body_side": "patient_right|midline|patient_left",
  "craniocaudal_level": "pelvic|lower_abdominal|upper_abdominal|thoracic",
  "anteroposterior_position": "posterior|central|anterior",
  "relative_size": "small|medium|large",
  "elongation": "round|oval|elongated",
  "contour_lobulation": "smooth|mildly_lobulated|markedly_irregular",
  "hilar_notch": "absent|subtle|prominent",
  "relative_attenuation": "much_darker|slightly_darker|similar|brighter|much_brighter",
  "interior_homogeneity": "homogeneous|mildly_heterogeneous|markedly_heterogeneous",
  "bright_cortical_rim": "absent|partial|complete",
  "boundary_definition": "indistinct|moderate|sharp",
  "adjacent_bone": "absent|small|prominent"
}
```

Zero-shot prompt (arm A). SHA-256 4e4c0e26b60ac15251411c5c98f6621f...

You are an expert reader of medical images. You answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 1-channel greyscale abdominal CT crops at a fixed abdominal window; each image is a 2D bounding-box crop around one organ taken from a CT volume in the axial plane. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 11 categories:

1. bladder
2. femur-left
3. femur-right
4. heart
5. kidney-left
6. kidney-right
7. liver
8. lung-left
9. lung-right
10. pancreas
11. spleen

How likely is each category for this image? Give every category a probability between 0 and 1, and make the 11 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 11 keys, the category names above copied exactly as written, each value a number.

```
{
  "bladder": <probability>,
  "femur-left": <probability>,
  "femur-right": <probability>,
  "heart": <probability>,
  "kidney-left": <probability>,
  "kidney-right": <probability>,
  "liver": <probability>,
  "lung-left": <probability>,
  "lung-right": <probability>,
  "pancreas": <probability>,
  "spleen": <probability>
}
```

A.7 breastmnist

Concept bank breastmnist.yaml, SHA-256 b5a6520b745334a4.... 7 concepts, 2 classes.

Concept prompt (arms B, C and C+P). SHA-256 ca722c5882b176037ccdfc95c6e4b5d7...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 1-channel greyscale breast ultrasound images. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 7 questions below about what is visible in this image. Each question lists its

levels in order. For every question choose the one level that best fits this image.

1. focal_lesion_present

Is a discrete focal area standing out from the surrounding layered tissue visible?

Levels, in order, and what each looks like:

absent: only smooth parallel layers of fat, fibroglandular tissue and muscle, with nothing standing out from them

equivocal: a faint textural variation whose outline cannot be traced all the way round on this single view

present: a discrete area whose full outline can be traced, occupying space and pushing the layers aside

2. shape_irregularity

If a focal area stands out, is its outline oval, gently lobulated, or jagged?

Levels, in order, and what each looks like:

no_focal_area: no traceable outline anywhere in the frame; the fibroglandular layers run uninterrupted across it

oval_or_round: a smooth ellipse or circle whose outline curves evenly, with no bulges and no corners

gently_lobulated: two or three broad rounded bulges on an otherwise smooth ellipse, each bulge wide and shallow

irregular: outline neither round nor oval: jagged or angular, or studded with many tiny 1-2 mm lobulations

3. margin_circumscription

If a focal area stands out, is its boundary sharp, blurred, or angular and spiky?

Levels, in order, and what each looks like:

no_focal_area: nothing stands out from the tissue layers, so there is no boundary anywhere to judge

circumscribed: the whole circumference is a sharp line with an abrupt transition to the tissue around it

indistinct: part of the edge fades into the surrounding tissue with no sharp line separating the two

angular_or_spiculated: the edge forms sharp corners, or fine straight lines radiate outward from it into the tissue

4. taller_than_wide

Does any focal area reach deeper below the skin than it measures across?

Levels, in order, and what each looks like:

absent: width across the image clearly exceeds depth, or nothing discrete stands out to measure at all

borderline: depth below the skin line and width across it look about equal, neither axis clearly the longer

present: the area reaches further in depth from the skin line than it measures across, standing upright

5. posterior_shadowing

Is tissue directly beneath a focal area darker than tissue beside it at that depth?

Levels, in order, and what each looks like:

absent: tissue just deep to the area is as bright as tissue at the same depth beside it
mild: narrow dark lines trail from the area's edges only; tissue directly behind it stays visible

marked: a broad dark band deep to the whole area hides the tissue behind it toward the image bottom

6. `thick_echogenic_halo`
 Is a broad bright rim, thicker than a fine line, present around a focal area?
 Levels, in order, and what each looks like:
 absent: no bright rim, or only a hairline bright capsule no thicker than the
 surrounding tissue lines
 mild: a bright rim clearly thicker than a hairline but present around only part of the
 circumference
 marked: a broad bright zone right around the area, its outer edge fading gradually into
 surrounding tissue

7. `tissue_plane_distortion`
 Are the horizontal tissue layers around a focal area pulled in or interrupted?
 Levels, in order, and what each looks like:
 absent: the horizontal tissue layers run straight past the area, neither bent toward it
 nor broken
 mild: layers next to the area bow or angle toward it but remain continuous along their
 length
 marked: layers are broken where they meet the area, with straight strands running
 radially out from it

Reply with one JSON object and nothing else: exactly 7 keys, the question names above, each
 value one of that question's levels copied exactly as written.

```
{
  "focal_lesion_present": "absent|equivocal|present",
  "shape_irregularity": "no_focal_area|oval_or_round|gently_lobulated|irregular",
  "margin_circumscription": "no_focal_area|circumscribed|indistinct|angular_or_spiculated",
  "taller_than_wide": "absent|borderline|present",
  "posterior_shadowing": "absent|mild|marked",
  "thick_echogenic_halo": "absent|mild|marked",
  "tissue_plane_distortion": "absent|mild|marked"
}
```

Zero-shot prompt (arm A). SHA-256 d21520d2fd08003306e3b719de057001...

You are an expert reader of medical images. You answer with the JSON object you are asked
 for and nothing else.

You are shown one image from a collection of 1-channel greyscale breast ultrasound images.
 The image is 224 by 224 pixels, and comes with no clinical context, no patient information
 and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 2 categories:

1. malignant
2. normal, benign

How likely is each category for this image? Give every category a probability between 0 and
 1, and make the 2 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 2 keys, the category names above copied
 exactly as written, each value a number.

```
{
```

```

    "malignant": <probability>,
    "normal, benign": <probability>
}

```

A.8 retinamnist

Concept bank retinamnist.yaml, SHA-256 9797b12c42d4254b... 10 concepts, 5 classes.

Concept prompt (arms B, C and C+P). SHA-256 3d5e29b1e6101da5ab273cbdbeb01b42...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 3-channel colour retinal fundus photographs. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 10 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. microaneurysms_dot_haemorrhages

How many small, sharply outlined round red dots, each far smaller than the width of a main retinal vessel, are scattered over the orange retinal background?

Levels, in order, and what each looks like:

none: no red dots anywhere on the orange background; the only red structures are the branching vessels themselves

few: a handful of red dots, roughly one to five, all lying within one part of the visible retina

moderate: red dots easily found in more than one quarter of the retina but still countable, roughly six to twenty in all

many: red dots in every quarter of the retina and too many to count singly, more than about twenty per quarter

2. blot_haemorrhages

How many larger red patches with blurred or irregular outlines, clearly bigger than the small round dots, are visible in the retina?

Levels, in order, and what each looks like:

none: no red patch is larger or blurrier than the fine round dots; the retina shows dots or nothing at all

few: one to about five blurred red patches, confined to one quarter of the visible retina

moderate: blurred red patches in two or three quarters of the retina, fewer than about twenty in any one of them

many: more than about twenty blurred red patches in each of the four quarters of the retina

3. hard_exudates

How much sharply outlined bright yellow-white deposit, sometimes in clusters, lines or rings, is visible in the retina?

Levels, in order, and what each looks like:

none: no yellow-white deposit on the retina; the brightest structure in the frame is the pale optic disc itself

- few: one or two small yellow-white specks or a single short line, together well under one optic-disc area
 moderate: several sharp-edged yellow-white clusters or a partial ring, together about one optic-disc area
 many: yellow-white deposit in rings or confluent sheets covering more than about two optic-disc areas
4. cotton_wool_spots
 How many pale greyish-white patches with soft, feathery, ill-defined edges are visible, distinct from the sharply outlined yellow deposits?
 Levels, in order, and what each looks like:
 none: no soft-edged pale patch; any bright lesion present has a crisp yellow border rather than a fuzzy white one
 few: anything from one up to about five fluffy white patches with feathery margins, however widely they are scattered
 many: more than about five fluffy white patches with feathery margins, too many to count on one hand
5. venous_beadings
 Do the larger dark retinal veins vary in calibre along their course, looking like a string of sausages rather than a smoothly tapering tube?
 Levels, in order, and what each looks like:
 absent: every large vein tapers smoothly from the disc outwards with no local widening or pinching
 mild: one vein shows a short stretch of alternating widening and narrowing, in one quarter of the retina only
 marked: definite sausage-string calibre change in veins in two or more of the four quarters of the retina
6. irma
 Are there fine, irregular, tortuous vessel segments lying within the retina, often near red lesions, that do not follow the orderly branching of the main vessels?
 Levels, in order, and what each looks like:
 absent: all visible vessels branch and taper in orderly fashion; no tangled segment sits between them
 mild: a faint or doubtful tangle of fine intraretinal vessels, hard to tell from a normal small branch
 marked: prominent tangles of fine irregular intraretinal vessels in one or more of the four quarters of the retina
7. lesion_quadrant_spread
 Taking the optic disc as the centre, in how many quadrants of the visible retina are red lesions present?
 Levels, in order, and what each looks like:
 none: no red dot or patch in any quarter of the retina that the frame actually shows
 one_quadrant: red lesions in one quarter only, quartering about the disc or, if it is off-frame, the image centre
 two_or_three_quadrants: red lesions in two or three of the four quarters, with at least one quarter free of them
 four_quadrants: red lesions in every one of the four quarters of the visible retina, none spared
8. neovascularisation
 Is there a fine net, tuft or frond of thin new vessels on or beside the optic disc, or

elsewhere along the vascular arcades, distinct from the normal branching vessels?
Levels, in order, and what each looks like:
absent: every vessel runs out from the disc along the arcades and tapers; no extra net lies on the disc or retinal surface
present: a fine tangled net, tuft or frond of thin vessels lies on the disc or the retinal surface, finer than the arcade it sits on

9. preretinal_or_vitreous_haemorrhage

Is there a dense sheet or pool of blood lying in front of the retina that hides the vessels beneath it, or a diffuse red haze obscuring part of the fundus?

Levels, in order, and what each looks like:

absent: the retinal vessels can be followed across the whole frame; nothing dense lies in front of them

present: a dense blood sheet, often with a flat upper level, hides the vessels under it, or a red haze veils part of the view

10. fibrous_proliferation

Is there white or grey fibrous, sheet-like or membranous tissue lying on the retinal surface, often over the disc or along the vascular arcades?

Levels, in order, and what each looks like:

absent: the retinal surface shows only vessels and orange background, with no opaque sheet lying over it

present: an opaque white or grey membranous sheet lies on the retinal surface, often bridging the disc and the arcades

Reply with one JSON object and nothing else: exactly 10 keys, the question names above, each value one of that question's levels copied exactly as written.

```
{
  "microaneurysms_dot_haemorrhages": "none|few|moderate|many",
  "blot_haemorrhages": "none|few|moderate|many",
  "hard_exudates": "none|few|moderate|many",
  "cotton_wool_spots": "none|few|many",
  "venous_beading": "absent|mild|marked",
  "irma": "absent|mild|marked",
  "lesion_quadrant_spread": "none|one_quadrant|two_or_three_quadrants|four_quadrants",
  "neovascularisation": "absent|present",
  "preretinal_or_vitreous_haemorrhage": "absent|present",
  "fibrous_proliferation": "absent|present"
}
```

Zero-shot prompt (arm A). SHA-256 fcff0f33d29a1d9df47446c8ffc67deb...

You are an expert reader of medical images. You answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 3-channel colour retinal fundus photographs. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 5 categories:

1. 0 (no apparent diabetic retinopathy)
2. 1 (mild non-proliferative diabetic retinopathy)
3. 2 (moderate non-proliferative diabetic retinopathy)

4. 3 (severe non-proliferative diabetic retinopathy)
5. 4 (proliferative diabetic retinopathy)

How likely is each category for this image? Give every category a probability between 0 and 1, and make the 5 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 5 keys, the category names above copied exactly as written, each value a number.

```
{
  "0": <probability>,
  "1": <probability>,
  "2": <probability>,
  "3": <probability>,
  "4": <probability>
}
```

A.9 tissuemnist

Concept bank tissuemnist.yaml, SHA-256 d352b03a30d3faa0.... 8 concepts, 8 classes.

Concept prompt (arms B, C and C+P). SHA-256 f18c15764a8dcd3fd0e552dc55846206...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of greyscale 2D projections of segmented single cells from 3D confocal fluorescence microscopy of nuclear-stained human kidney cortex. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 8 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. nucleus_size

How large is the single bright object at the centre of the image compared with the width of the whole image?

Levels, in order, and what each looks like:

small: occupies less than about a third of the frame width, with wide dark margins on every side

medium: spans roughly half the frame width, the crop being about two nucleus-widths across

large: spans more than about two thirds of the frame width and nearly reaches opposite edges

2. nucleus_elongation

Is the central bright object about as wide as it is tall, moderately longer in one direction, or clearly long and narrow?

Levels, in order, and what each looks like:

round: about as wide as it is tall; no direction is clearly the longer one

oval: clearly longer one way than the other, but still blunt and rounded at both ends

elongated: a narrow strip whose long axis runs far across the frame while staying thin

along its whole length

3. outline_regularity

Is the outline of the central bright object smooth and unbroken, or is it notched, indented, or split into separate lobes?

Levels, in order, and what each looks like:

smooth: the bright edge runs unbroken all the way round, with no notch cutting into it
indented: one or two dark notches bite into the edge without dividing the object in two
lobed: two or more bright blobs joined by thin bridges; a neutrophil nucleus carries three to five such lobes

4. chromatin_texture

Inside the central bright object, is the signal even across it, or is it broken up into brighter and darker patches?

Levels, in order, and what each looks like:

uniform: brightness is even across the object; no patch stands out against its surroundings
mottled: faint lighter and darker patches show, but they blend into each other without sharp edges
coarsely_clumped: a few distinctly bright blocks sit against clearly darker gaps, with abrupt borders between them

5. staining_intensity

How bright is the central object compared with the darkest background in the image?

Levels, in order, and what each looks like:

dim: grey rather than white, only modestly above the dark background even at its centre
moderate: solidly bright over most of its area, with only the middle approaching white
bright: close to white and near-saturated over most of its area, the strongest signal in the frame

6. neighbouring_nuclei

Apart from the central object, how many other bright objects or parts of them are visible anywhere in the image, including at the edges?

Levels, in order, and what each looks like:

none: the central object is the only bright thing; all four frame edges stay dark
one_or_two: one or two further bright areas appear, typically cut off by the frame edge
several: three or more separate bright areas, so several frame edges carry a partial nucleus

7. neighbour_variety

Do the other bright objects in the image look like near-copies of one another in size and shape, or a mixture of clearly different sizes and shapes?

Levels, in order, and what each looks like:

uniform: the other bright areas match one another closely in both size and shape
somewhat_mixed: the other bright areas differ noticeably in size or in shape, but not in both
markedly_mixed: the other bright areas differ in size and in shape at once: round, narrow and broad ones together

8. size_versus_neighbours

Compared with the other bright objects visible in the image, is the central one smaller, about the same size, or larger?

Levels, in order, and what each looks like:

smaller: visibly narrower than the partial bright areas showing at the frame edge

similar: the same width as the partial nuclei at the frame edge, as far as one can tell
larger: clearly wider than any partial nucleus visible at the frame edge

Reply with one JSON object and nothing else: exactly 8 keys, the question names above, each value one of that question's levels copied exactly as written.

```
{
  "nucleus_size": "small|medium|large",
  "nucleus_elongation": "round|oval|elongated",
  "outline_regularity": "smooth|indented|lobed",
  "chromatin_texture": "uniform|mottled|coarsely_clumped",
  "staining_intensity": "dim|moderate|bright",
  "neighbouring_nuclei": "none|one_or_two|several",
  "neighbour_variety": "uniform|somewhat_mixed|markedly_mixed",
  "size_versus_neighbours": "smaller|similar|larger"
}
```

Zero-shot prompt (arm A). SHA-256 ae4e7be8c9b399aa9b8a00f479d48d9b...

You are an expert reader of medical images. You answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of greyscale 2D projections of segmented single cells from 3D confocal fluorescence microscopy of nuclear-stained human kidney cortex. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 8 categories:

1. Collecting Duct, Connecting Tubule
2. Distal Convoluted Tubule
3. Glomerular endothelial cells
4. Interstitial endothelial cells
5. Leukocytes
6. Podocytes
7. Proximal Tubule Segments
8. Thick Ascending Limb

How likely is each category for this image? Give every category a probability between 0 and 1, and make the 8 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 8 keys, the category names above copied exactly as written, each value a number.

```
{
  "Collecting Duct, Connecting Tubule": <probability>,
  "Distal Convoluted Tubule": <probability>,
  "Glomerular endothelial cells": <probability>,
  "Interstitial endothelial cells": <probability>,
  "Leukocytes": <probability>,
  "Podocytes": <probability>,
  "Proximal Tubule Segments": <probability>,
  "Thick Ascending Limb": <probability>
}
```

A.10 organcmnist

Concept bank organcmnist.yaml, SHA-256 219a94de93aa9bb7... 12 concepts, 11 classes.

Concept prompt (arms B, C and C+P). SHA-256 66a341b936e8ba9511bcf3f03a6ecacb...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 1-channel greyscale abdominal CT crops at a fixed abdominal window; each image is a 2D bounding-box crop around one organ taken from a CT volume in the coronal plane. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 12 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. body_side

Judging from the surrounding structures caught in the crop (spine, body wall, midline vessels), does the main structure lie on the patient's right, near the midline, or on the patient's left? Remember that in a cross-sectional image the patient's left is shown on the right of the picture.

Levels, in order, and what each looks like:

patient_right: the spine or the aorta sits toward the image-right of the crop, with the structure lying to the left of it on screen

midline: body wall or paired muscles lie about equally far from the structure on both sides of the image, with no side landmark nearer than the other

patient_left: the spine or the aorta sits toward the image-left of the crop, with the structure lying to the right of it on screen

2. craniocaudal_level

From the surroundings visible in the crop, at what level of the trunk was this section taken: the pelvis, the lower abdomen, the upper abdomen, or the chest?

Levels, in order, and what each looks like:

pelvic: an iliac wing, pubic bone or rounded femoral head borders the crop, and no rib is present

lower_abdominal: gas- or contrast-filled bowel loops and psoas muscle border the crop, with neither rib nor pelvic bone in view

upper_abdominal: the lowest ribs border the crop alongside solid parenchymal organs, with no pelvic bone in view

thoracic: ribs surround the crop and nearly black air-filled lung borders the structure

3. anteroposterior_position

Do the surroundings place the main structure toward the back of the body (against the spine and the back muscles), in the middle, or toward the front body wall?

Levels, in order, and what each looks like:

posterior: a vertebral body and the thick paraspinal back muscles lie immediately against the structure

central: neither the spine nor the front body wall touches the structure; both lie beyond it at the crop edge

anterior: the front body wall, with its strap-like rectus muscle or the sternum, lies immediately against the structure

4. relative_size

Compared with the landmarks caught at the edge of the crop (a vertebral body, a rib, the thickness of the body wall), is the main structure small, medium or large?

Levels, in order, and what each looks like:

small: a single landmark at the crop edge - one vertebra or one rib - appears magnified and fills much of that edge

medium: two or three separate landmarks fit along the crop edge, each still plainly resolved

large: many small landmarks crowd the crop edge at once - several ribs, vertebrae or bowel loops together

5. elongation

Is the outline of the main structure roughly circular, moderately oval, or markedly elongated and strip-like?

Levels, in order, and what each looks like:

round: the two widths of the outline differ by less than about a quarter, and the shape fills most of the frame

oval: the long width is roughly one and a half to three times the short one, leaving the frame corners empty

elongated: the long axis runs more than about three times the short one, a strip that leaves most of the frame as background

6. contour_lobulation

Is the outline smooth and even, mildly lobulated or wavy, or markedly irregular and feathery?

Levels, in order, and what each looks like:

smooth: the border runs as an unbroken curve, with no bump deeper than the thickness of the border line itself

mildly_lobulated: one or two shallow bulges or gentle waves interrupt an otherwise continuous curve

markedly_irregular: many bumps and clefts break the border, or it dissolves into a feathery fringe against the surrounding fat

7. hilar_notch

Is there a single distinct indentation, notch or wedge cut into one side of the outline where vessels appear to enter?

Levels, in order, and what each looks like:

absent: the border runs continuously round the structure with no indentation on any side

subtle: one side is flattened or dimpled, but the dent is shallower than about a tenth of the structure's width

prominent: a wedge deeper than about a tenth of the width is cut into one side, with vessels or fat running into it

8. relative_attenuation

Overall, how bright is the main structure compared with the ordinary soft tissue and muscle visible around it?

Levels, in order, and what each looks like:

much_darker: nearly black at this window, clearly the darkest tissue in the crop and far below the muscle at the frame edge

slightly_darker: a shade below the muscle at the frame edge, but plainly mid-grey rather than black

similar: no step in grey level against the muscle at the frame edge; only the outline

separates the two
brighter: a clear step above the muscle at the frame edge, yet still well short of the white of bone
much_brighter: among the brightest things in the crop, approaching the white of cortical bone or of dense contrast

9. interior_homogeneity

Is the inside of the structure uniform in brightness, or does it contain clearly different regions, internal branching lines or pockets?

Levels, in order, and what each looks like:

homogeneous: one even grey fills the interior, broken by nothing more than a few thin vessel lines

mildly_heterogeneous: faint mottling or a few branching lines cross the interior, but no region differs sharply from the rest

markedly_heterogeneous: two or more distinct zones, one clearly brighter than the other, meet at sharp internal borders

10. bright_cortical_rim

Is the structure outlined by a thin rim of the very brightest, calcium-density material, brighter than any soft tissue in the crop?

Levels, in order, and what each looks like:

absent: no calcium-white line follows the border anywhere; the edge is grey meeting grey

partial: a calcium-white line brighter than any soft tissue follows part of the border and then breaks off

complete: an unbroken calcium-white line rings the whole border, the brightest thing in the crop

11. boundary_definition

How clearly is the edge of the main structure separated from what surrounds it: barely distinguishable, moderately defined, or crisp along most of its circumference?

Levels, in order, and what each looks like:

indistinct: the edge merges into the surrounding tissue over most of its length; you could not trace it confidently

moderate: the edge is traceable over roughly half the circumference and blurred or interrupted over the rest

sharp: a crisp step in brightness follows almost the whole circumference, often with a dark fat line beside it

12. adjacent_bone

Are very bright, calcium-density structures such as vertebrae, ribs or pelvic bone visible at the edge of the crop, and if so how much of the edge do they occupy?

Levels, in order, and what each looks like:

absent: nothing of calcium-white density appears anywhere along the crop edge

small: one small patch of calcium-white touches the crop edge and occupies a minor part of it

prominent: calcium-white runs along a long stretch of the crop edge, or several bony pieces border the crop at once

Reply with one JSON object and nothing else: exactly 12 keys, the question names above, each value one of that question's levels copied exactly as written.

```
{  
  "body_side": "patient_right|midline|patient_left",
```

```

"craniocaudal_level": "pelvic|lower_abdominal|upper_abdominal|thoracic",
"anteroposterior_position": "posterior|central|anterior",
"relative_size": "small|medium|large",
"elongation": "round|oval|elongated",
"contour_lobulation": "smooth|mildly_lobulated|markedly_irregular",
"hilar_notch": "absent|subtle|prominent",
"relative_attenuation": "much_darker|slightly_darker|similar|brighter|much_brighter",
"interior_homogeneity": "homogeneous|mildly_heterogeneous|markedly_heterogeneous",
"bright_cortical_rim": "absent|partial|complete",
"boundary_definition": "indistinct|moderate|sharp",
"adjacent_bone": "absent|small|prominent"
}

```

Zero-shot prompt (arm A). SHA-256 769cc6a4ea3b8c19e7ba8c780bdb0a12...

You are an expert reader of medical images. You answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 1-channel greyscale abdominal CT crops at a fixed abdominal window; each image is a 2D bounding-box crop around one organ taken from a CT volume in the coronal plane. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 11 categories:

1. bladder
2. femur-left
3. femur-right
4. heart
5. kidney-left
6. kidney-right
7. liver
8. lung-left
9. lung-right
10. pancreas
11. spleen

How likely is each category for this image? Give every category a probability between 0 and 1, and make the 11 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 11 keys, the category names above copied exactly as written, each value a number.

```

{
  "bladder": <probability>,
  "femur-left": <probability>,
  "femur-right": <probability>,
  "heart": <probability>,
  "kidney-left": <probability>,
  "kidney-right": <probability>,
  "liver": <probability>,
  "lung-left": <probability>,
  "lung-right": <probability>,
  "pancreas": <probability>,

```

```
"spleen": <probability>
}
```

A.11 organsmnist

Concept bank organsmnist.yaml, SHA-256 e39a65050100b4ff... 12 concepts, 11 classes.

Concept prompt (arms B, C and C+P). SHA-256 35223f93915ed25b138a6b77c2f34637...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 1-channel greyscale abdominal CT crops at a fixed abdominal window; each image is a 2D bounding-box crop around one organ taken from a CT volume in the sagittal plane. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 12 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. body_side

Judging from the surrounding structures caught in the crop (spine, body wall, midline vessels), does the main structure lie on the patient's right, near the midline, or on the patient's left? Remember that in a cross-sectional image the patient's left is shown on the right of the picture.

Levels, in order, and what each looks like:

patient_right: the spine or the aorta sits toward the image-right of the crop, with the structure lying to the left of it on screen

midline: body wall or paired muscles lie about equally far from the structure on both sides of the image, with no side landmark nearer than the other

patient_left: the spine or the aorta sits toward the image-left of the crop, with the structure lying to the right of it on screen

2. craniocaudal_level

From the surroundings visible in the crop, at what level of the trunk was this section taken: the pelvis, the lower abdomen, the upper abdomen, or the chest?

Levels, in order, and what each looks like:

pelvic: an iliac wing, pubic bone or rounded femoral head borders the crop, and no rib is present

lower_abdominal: gas- or contrast-filled bowel loops and psoas muscle border the crop, with neither rib nor pelvic bone in view

upper_abdominal: the lowest ribs border the crop alongside solid parenchymal organs, with no pelvic bone in view

thoracic: ribs surround the crop and nearly black air-filled lung borders the structure

3. anteroposterior_position

Do the surroundings place the main structure toward the back of the body (against the spine and the back muscles), in the middle, or toward the front body wall?

Levels, in order, and what each looks like:

posterior: a vertebral body and the thick paraspinal back muscles lie immediately against the structure

central: neither the spine nor the front body wall touches the structure; both lie beyond it at the crop edge
anterior: the front body wall, with its strap-like rectus muscle or the sternum, lies immediately against the structure

4. relative_size

Compared with the landmarks caught at the edge of the crop (a vertebral body, a rib, the thickness of the body wall), is the main structure small, medium or large?

Levels, in order, and what each looks like:

small: a single landmark at the crop edge - one vertebra or one rib - appears magnified and fills much of that edge

medium: two or three separate landmarks fit along the crop edge, each still plainly resolved

large: many small landmarks crowd the crop edge at once - several ribs, vertebrae or bowel loops together

5. elongation

Is the outline of the main structure roughly circular, moderately oval, or markedly elongated and strip-like?

Levels, in order, and what each looks like:

round: the two widths of the outline differ by less than about a quarter, and the shape fills most of the frame

oval: the long width is roughly one and a half to three times the short one, leaving the frame corners empty

elongated: the long axis runs more than about three times the short one, a strip that leaves most of the frame as background

6. contour_lobulation

Is the outline smooth and even, mildly lobulated or wavy, or markedly irregular and feathery?

Levels, in order, and what each looks like:

smooth: the border runs as an unbroken curve, with no bump deeper than the thickness of the border line itself

mildly_lobulated: one or two shallow bulges or gentle waves interrupt an otherwise continuous curve

markedly_irregular: many bumps and clefts break the border, or it dissolves into a feathery fringe against the surrounding fat

7. hilar_notch

Is there a single distinct indentation, notch or wedge cut into one side of the outline where vessels appear to enter?

Levels, in order, and what each looks like:

absent: the border runs continuously round the structure with no indentation on any side

subtle: one side is flattened or dimpled, but the dent is shallower than about a tenth of the structure's width

prominent: a wedge deeper than about a tenth of the width is cut into one side, with vessels or fat running into it

8. relative_attenuation

Overall, how bright is the main structure compared with the ordinary soft tissue and muscle visible around it?

Levels, in order, and what each looks like:

much_darker: nearly black at this window, clearly the darkest tissue in the crop and

far below the muscle at the frame edge
slightly_darker: a shade below the muscle at the frame edge, but plainly mid-grey rather than black
similar: no step in grey level against the muscle at the frame edge; only the outline separates the two
brighter: a clear step above the muscle at the frame edge, yet still well short of the white of bone
much_brighter: among the brightest things in the crop, approaching the white of cortical bone or of dense contrast

9. interior_homogeneity

Is the inside of the structure uniform in brightness, or does it contain clearly different regions, internal branching lines or pockets?

Levels, in order, and what each looks like:

homogeneous: one even grey fills the interior, broken by nothing more than a few thin vessel lines

mildly_heterogeneous: faint mottling or a few branching lines cross the interior, but no region differs sharply from the rest

markedly_heterogeneous: two or more distinct zones, one clearly brighter than the other, meet at sharp internal borders

10. bright_cortical_rim

Is the structure outlined by a thin rim of the very brightest, calcium-density material, brighter than any soft tissue in the crop?

Levels, in order, and what each looks like:

absent: no calcium-white line follows the border anywhere; the edge is grey meeting grey

partial: a calcium-white line brighter than any soft tissue follows part of the border and then breaks off

complete: an unbroken calcium-white line rings the whole border, the brightest thing in the crop

11. boundary_definition

How clearly is the edge of the main structure separated from what surrounds it: barely distinguishable, moderately defined, or crisp along most of its circumference?

Levels, in order, and what each looks like:

indistinct: the edge merges into the surrounding tissue over most of its length; you could not trace it confidently

moderate: the edge is traceable over roughly half the circumference and blurred or interrupted over the rest

sharp: a crisp step in brightness follows almost the whole circumference, often with a dark fat line beside it

12. adjacent_bone

Are very bright, calcium-density structures such as vertebrae, ribs or pelvic bone visible at the edge of the crop, and if so how much of the edge do they occupy?

Levels, in order, and what each looks like:

absent: nothing of calcium-white density appears anywhere along the crop edge

small: one small patch of calcium-white touches the crop edge and occupies a minor part of it

prominent: calcium-white runs along a long stretch of the crop edge, or several bony pieces border the crop at once

Reply with one JSON object and nothing else: exactly 12 keys, the question names above, each

value one of that question's levels copied exactly as written.

```
{
  "body_side": "patient_right|midline|patient_left",
  "craniocaudal_level": "pelvic|lower_abdominal|upper_abdominal|thoracic",
  "anteroposterior_position": "posterior|central|anterior",
  "relative_size": "small|medium|large",
  "elongation": "round|oval|elongated",
  "contour_lobulation": "smooth|mildly_lobulated|markedly_irregular",
  "hilar_notch": "absent|subtle|prominent",
  "relative_attenuation": "much_darker|slightly_darker|similar|brighter|much_brighter",
  "interior_homogeneity": "homogeneous|mildly_heterogeneous|markedly_heterogeneous",
  "bright_cortical_rim": "absent|partial|complete",
  "boundary_definition": "indistinct|moderate|sharp",
  "adjacent_bone": "absent|small|prominent"
}
```

Zero-shot prompt (arm A). SHA-256 9b5b8c195c1292816191891ffee0cab...

You are an expert reader of medical images. You answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of 1-channel greyscale abdominal CT crops at a fixed abdominal window; each image is a 2D bounding-box crop around one organ taken from a CT volume in the sagittal plane. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

The image belongs to exactly one of these 11 categories:

1. bladder
2. femur-left
3. femur-right
4. heart
5. kidney-left
6. kidney-right
7. liver
8. lung-left
9. lung-right
10. pancreas
11. spleen

How likely is each category for this image? Give every category a probability between 0 and 1, and make the 11 probabilities sum to 1.

Reply with one JSON object and nothing else: exactly 11 keys, the category names above copied exactly as written, each value a number.

```
{
  "bladder": <probability>,
  "femur-left": <probability>,
  "femur-right": <probability>,
  "heart": <probability>,
  "kidney-left": <probability>,
  "kidney-right": <probability>,
```

```

"liver": <probability>,
"lung-left": <probability>,
"lung-right": <probability>,
"pancreas": <probability>,
"spleen": <probability>
}

```

A.12 chestmnist

Concept bank chestmnist.yaml, SHA-256 adc12367b651d58c... 12 concepts, 14 findings.

Concept prompt (arms B, C and C+P). SHA-256 ea3b166cea4133d278d5b8edfaa65f21...

You are an expert reader of medical images. You report only what is visible in the image, and you answer with the JSON object you are asked for and nothing else.

You are shown one image from a collection of frontal chest radiographs, single-channel greyscale. The image is 224 by 224 pixels, and comes with no clinical context, no patient information and no scale bar: judge only what this image shows.

Answer all 12 questions below about what is visible in this image. Each question lists its levels in order. For every question choose the one level that best fits this image.

1. airspace_opacity

Is there a homogeneous white opacity filling part of the lung, dense enough that the vessels running through that region can no longer be traced?

Levels, in order, and what each looks like:

- absent: vessels can be followed from the hilum out to the periphery against dark lung, with no region of uniform added whiteness
- patchy: one or more ill-defined white areas, each under about a quarter of a lung, obscuring the margins of the vessels inside them
- confluent: one uniform white region obscuring vessel and airway margins across most of a lobe or a whole half of the chest

2. hazy_opacity

Is there a hazy, veil-like increase in whiteness of the lung through which the vessel and airway margins can still be seen?

Levels, in order, and what each looks like:

- absent: the lung is uniformly dark between the vessels, with no veil of added grey over any zone
- mild: a faint veil of grey over part of one lung through which vessel margins stay traceable, only less sharp
- marked: a dense veil over most of both lungs, vessels only just discernible through it but still not wholly obscured

3. interstitial_markings

How prominent are the fine linear, net-like markings of the lung, over and above the normal smoothly branching vessels?

Levels, in order, and what each looks like:

- normal: only smoothly branching vessels that taper outwards and fade before reaching the chest wall
- increased: thin lines persist out to the chest wall, including short horizontal lines

at the bases meeting the ribs at right angles
coarse_reticular: innumerable small linear opacities cross one another into a visible net, coarser and more angular than any vessel

4. opacity_basal_predominance

If any increased whiteness is present, how strongly is it concentrated in the lower zones near the diaphragm rather than spread evenly or set in the upper zones?

Levels, in order, and what each looks like:

absent: any added whiteness is spread evenly between upper and lower zones, or sits mainly above the hila

mild: the lower zones are visibly whiter than the upper zones, but the same change still reaches above the hila

marked: the whiteness is confined below the hila and the upper zones are as dark as normal lung

5. costophrenic_angle_blunting

Is the sharp angle where the diaphragm meets the chest wall blunted or filled in, with an upward-curving upper edge against the chest wall?

Levels, in order, and what each looks like:

absent: on both sides the diaphragm meets the chest wall in a sharp acute point

mild: the angle is rounded off and filled in, but the whole curve of the diaphragm dome is still visible

marked: whiteness fills the angle and climbs the chest wall with a concave upper edge, hiding part of the dome

6. peripheral_pleural_density

Is there a band or plaque of soft-tissue whiteness lying along the inner chest wall or capping a lung apex, following the surface of the lung rather than a vessel?

Levels, in order, and what each looks like:

absent: lung markings reach the ribs, with only rib shadows along the inner chest wall and no band of soft-tissue grey

mild: a thin grey band, or a cap over one lung apex, follows the chest wall and is narrower than one rib

marked: a rind of soft-tissue whiteness wider than a rib follows the chest wall over several rib spaces

7. cardiac_silhouette_width

How wide is the heart shadow compared with the widest inner diameter of the chest, a ratio of about one half being the usual upper limit?

Levels, in order, and what each looks like:

narrow: the heart shadow spans under about a third of the widest inner chest diameter and looks tall and vertical

normal: the heart shadow spans between about a third and one half of the widest inner chest diameter

wide: the heart shadow spans more than half of the widest inner chest diameter

8. discrete_rounded_opacity

Is there a discrete, roughly round, sharply outlined white lesion inside the lung, and if so is it small (under about one fifth the width of one lung, roughly 3 cm) or larger?

Levels, in order, and what each looks like:

absent: no rounded, sharply outlined white spot stands out from the branching pattern of the vessels

small: a rounded opacity up to about a fifth of one hemithorax wide (3 cm); at 1 cm it is near 6 pixels here and unreliable

large: a rounded opacity wider than about a fifth of one hemithorax, that is, over 3 cm across

9. pleural_line_with_avascular_zone

Is a thin white line visible running parallel to the chest wall, with completely blank lung beyond it and no vessel markings reaching out to the ribs?

Levels, in order, and what each looks like:

absent: vessel markings run out to the ribs all round, with no thin white line running parallel to the chest wall

present: a thin white line parallels the chest wall with blank, vessel-free lung between it and the ribs

10. lung_inflation

How inflated do the lungs look: short with high domed diaphragms, normal, or tall and dark with low flattened diaphragms and widely spaced ribs?

Levels, in order, and what each looks like:

reduced: the diaphragm domes sit high and steeply curved, the rib spaces are narrowed and the lungs look short

normal: gently domed diaphragms near the tenth posterior rib, with evenly and moderately spread rib spaces

increased: diaphragm outlines flat or concave upward, rib spaces widened, dark lung reaching below the tenth posterior rib

11. mediastinal_shift_direction

Is the trachea or the heart shadow off the midline, and if so is it pulled towards the whiter (more opaque) half of the chest or pushed away from it?

Levels, in order, and what each looks like:

towards_whiter_side: the tracheal air column and heart shadow are displaced into the whiter half of the chest, the darker half looking larger

none: the tracheal air column runs down the middle of the frame and the heart shadow sits centred between the lungs

away_from_whiter_side: the tracheal air column and heart shadow are pushed into the darker half, away from the whiter one

12. supradiaphragmatic_gas_structure

Is there an abnormal rounded structure projected behind the heart or above the diaphragm that contains gas or a gas-fluid level and distorts the diaphragm outline?

Levels, in order, and what each looks like:

absent: the diaphragm outline is a smooth unbroken curve with no gas pocket above it behind the heart

present: a rounded gas-containing pocket, sometimes with a flat fluid level, projects above the diaphragm behind the heart

Reply with one JSON object and nothing else: exactly 12 keys, the question names above, each value one of that question's levels copied exactly as written.

```
{
  "airspace_opacity": "absent|patchy|confluent",
  "hazy_opacity": "absent|mild|marked",
  "interstitial_markings": "normal|increased|coarse_reticular",
  "opacity_basal_predominance": "absent|mild|marked",
  "costophrenic_angle_blunting": "absent|mild|marked",
  "peripheral_pleural_density": "absent|mild|marked",
  "cardiac_silhouette_width": "narrow|normal|wide",
```

```

"discrete_rounded_opacity": "absent|small|large",
"pleural_line_with_avascular_zone": "absent|present",
"lung_inflation": "reduced|normal|increased",
"mediastinal_shift_direction": "towards_whiter_side|none|away_from_whiter_side",
"supradiaphragmatic_gas_structure": "absent|present"
}

```

Zero-shot prompt (arm A). Not rendered: chestmnist is multi-label, so arm A is not defined for it.

B Sampled images, by class

Every image below is one this study scored: the montages are drawn from the same cached sample arrays the vision–language model and the ImageNet encoder read, in sample order, so they are the evidence rather than an illustration of it. Each row is one class, labelled with how many of that class the seeded test sample happens to contain — which is where the thin-class caveat of the Data section becomes visible, and why a one-vs-rest column built on five positives carries the same weight in the AUC as one built on hundreds. A grey cell means the sample holds fewer images of that class than the six shown. For the multi-label task each row is a finding and its images carry other findings besides.

B.1 pathmnist

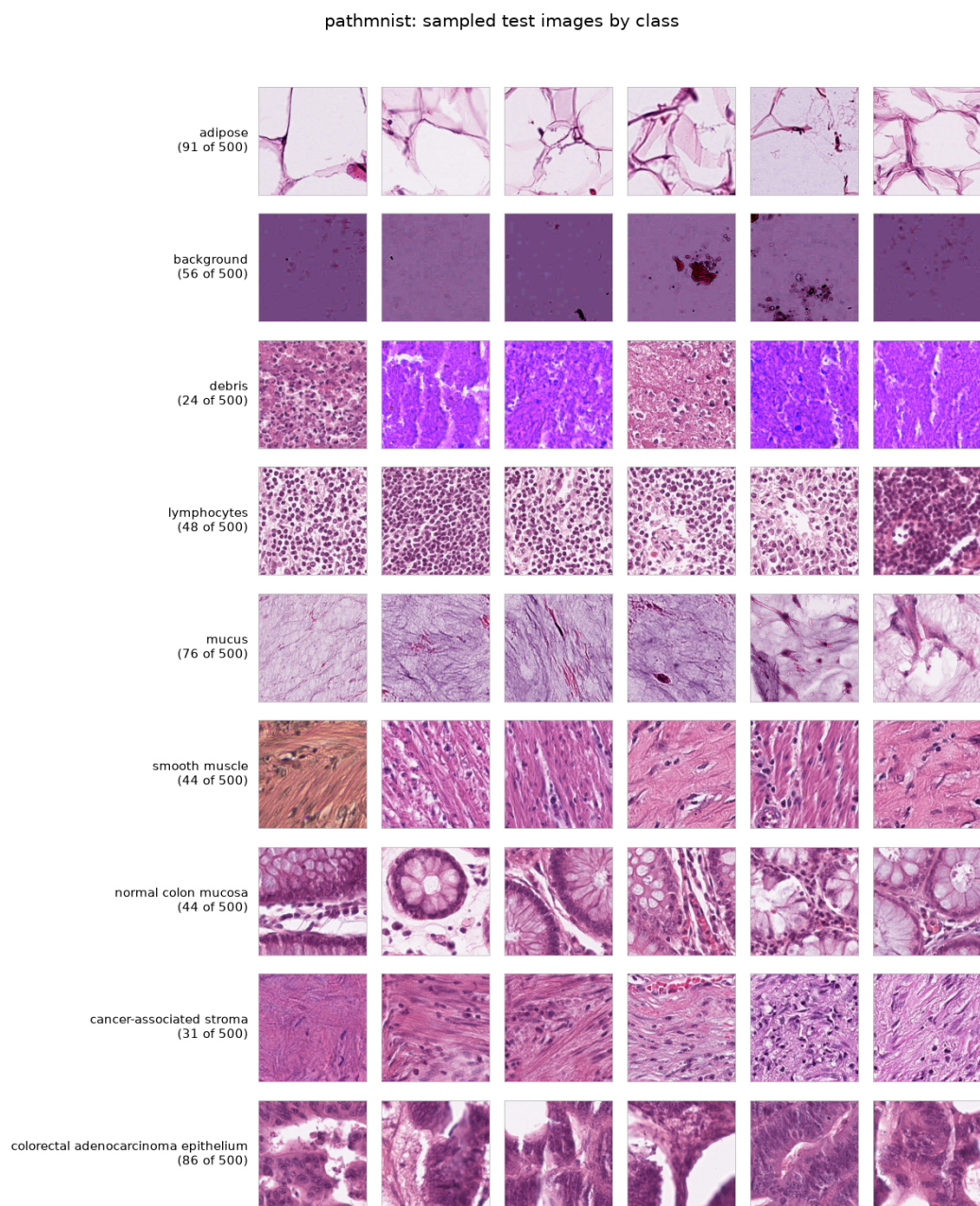


Figure 9: pathmnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.2 dermmnist

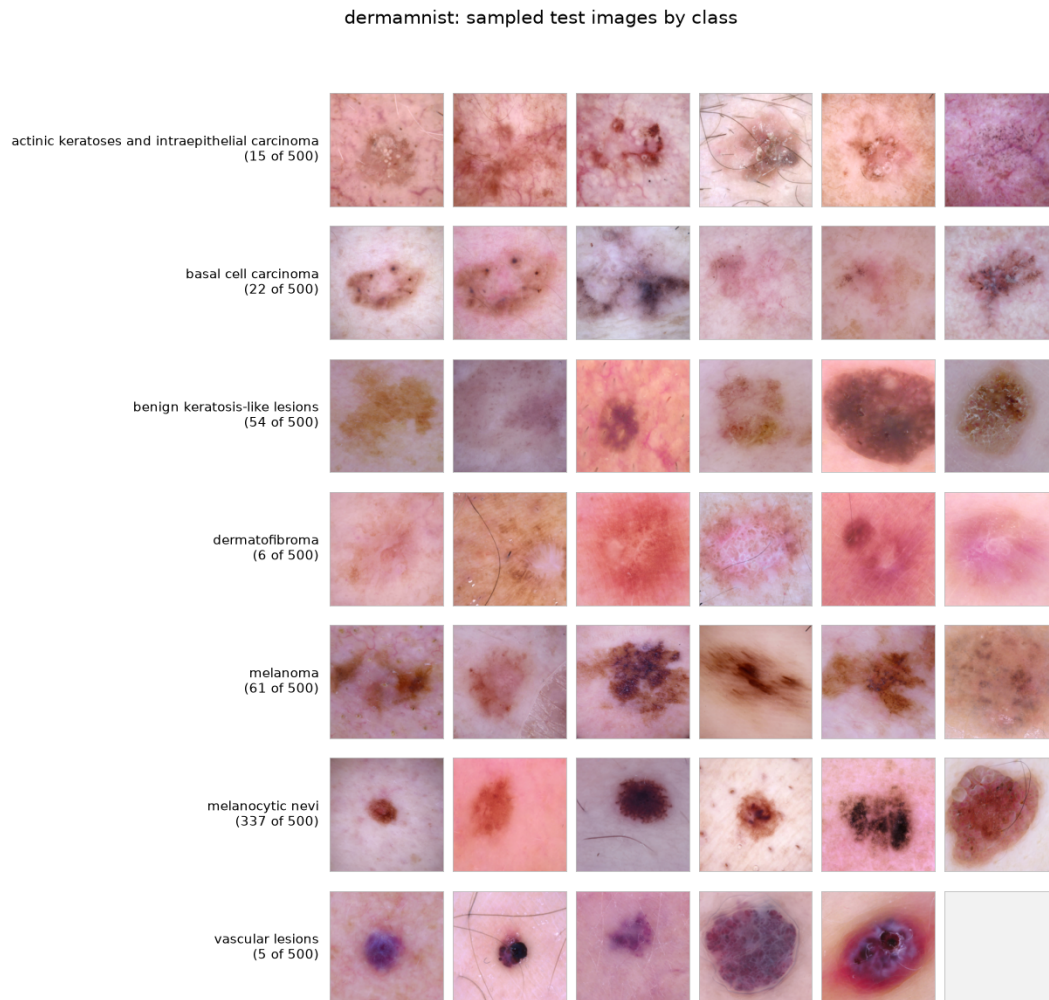


Figure 10: dermmnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.3 octmnist



Figure 11: octmnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.4 pneumoniarnist

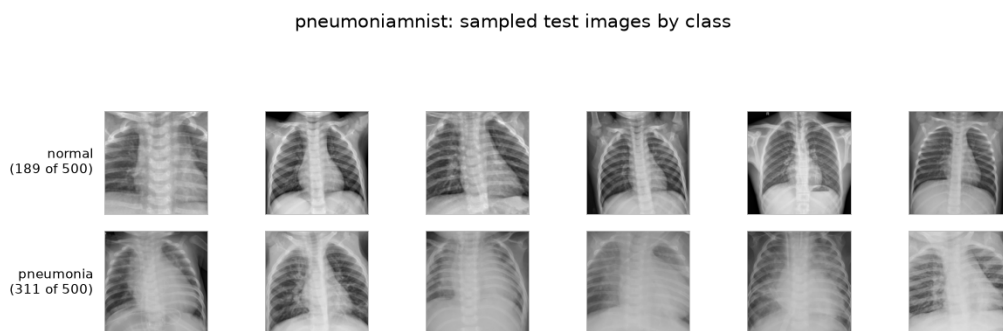


Figure 12: pneumoniarnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.5 bloodmnist

bloodmnist: sampled test images by class

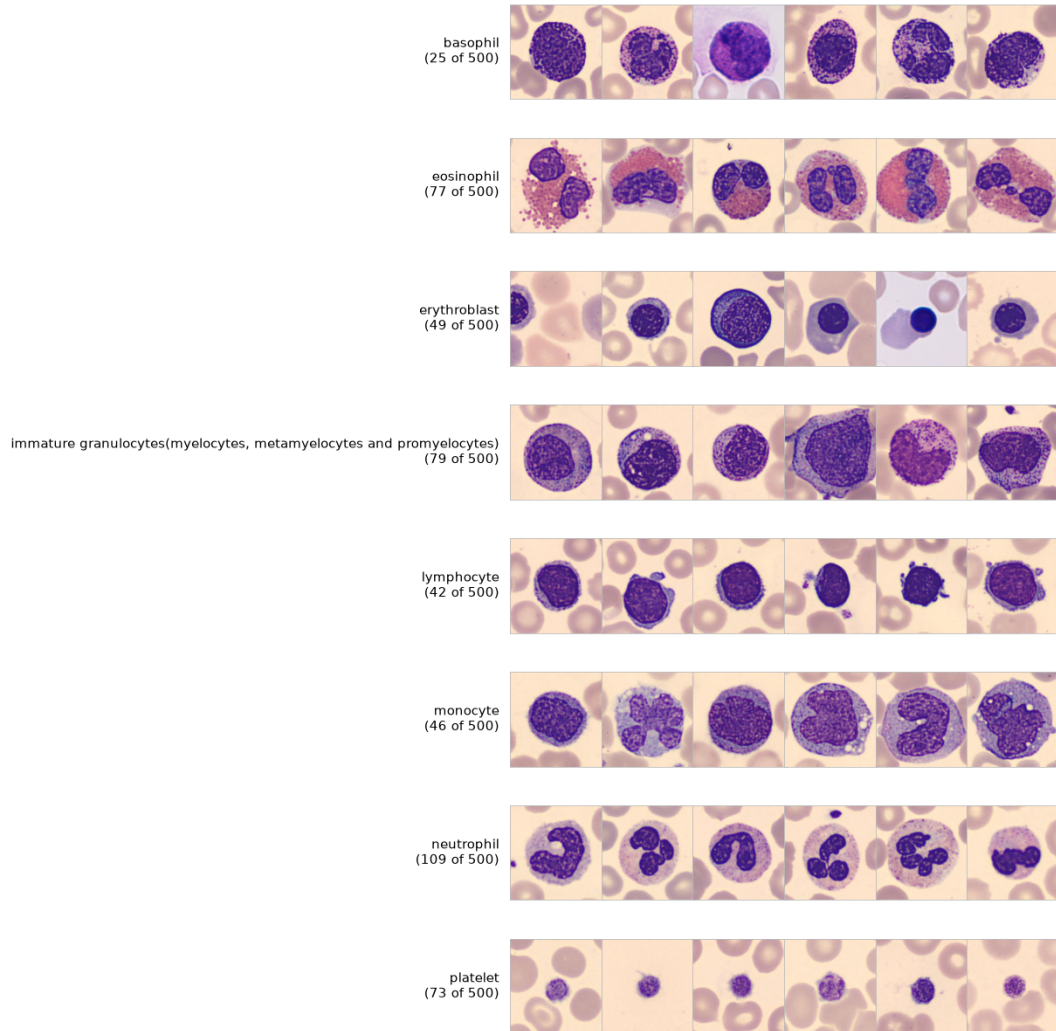


Figure 13: bloodmnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.6 organamnist

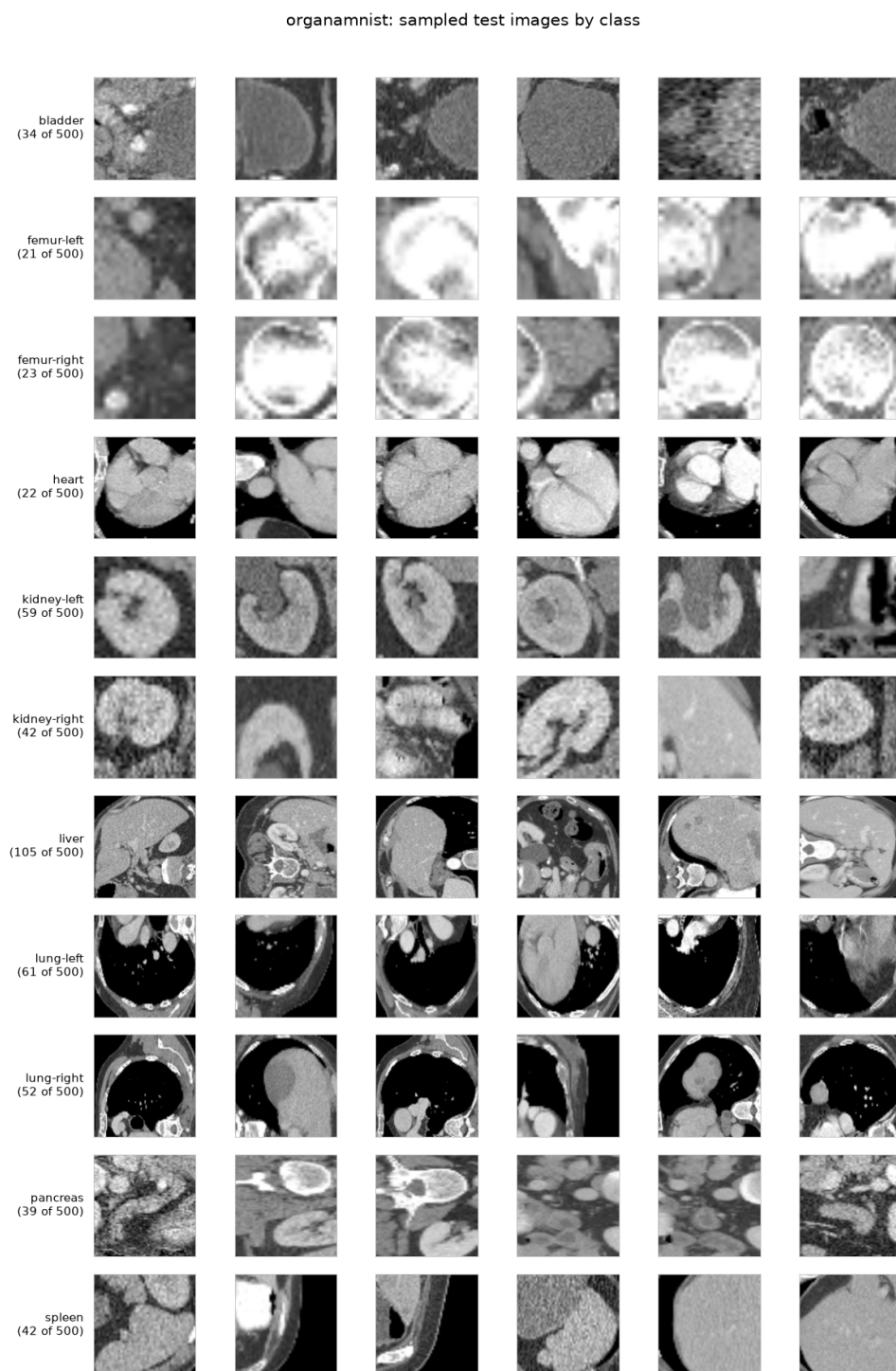


Figure 14: organamnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.7 breastmnist

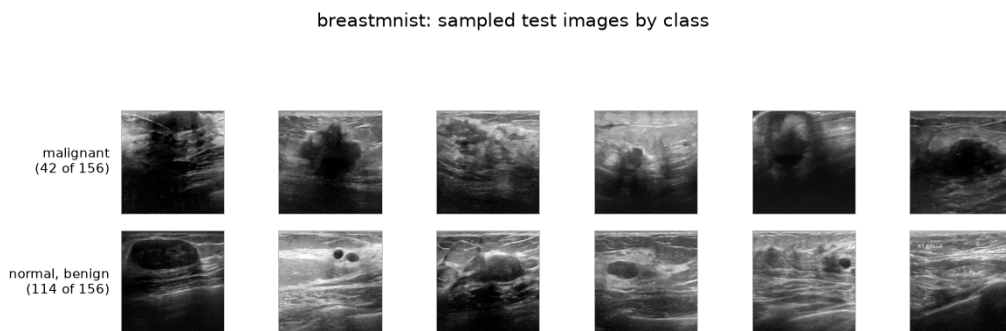


Figure 15: breastmnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.8 retinamnist

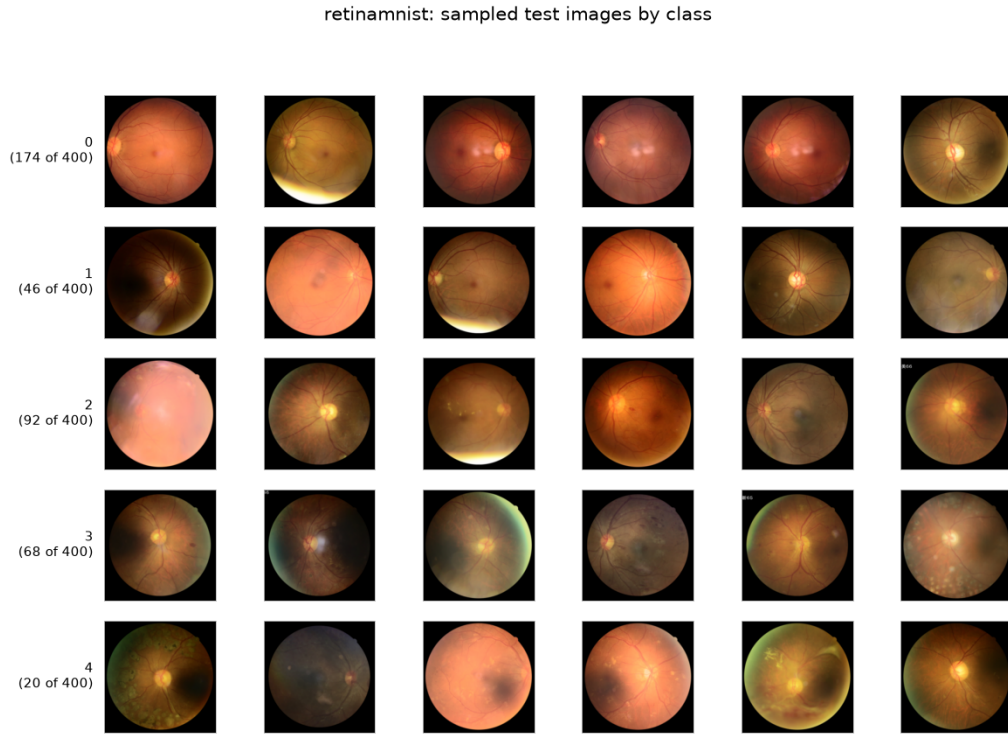


Figure 16: retinamnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.9 tissuemnist

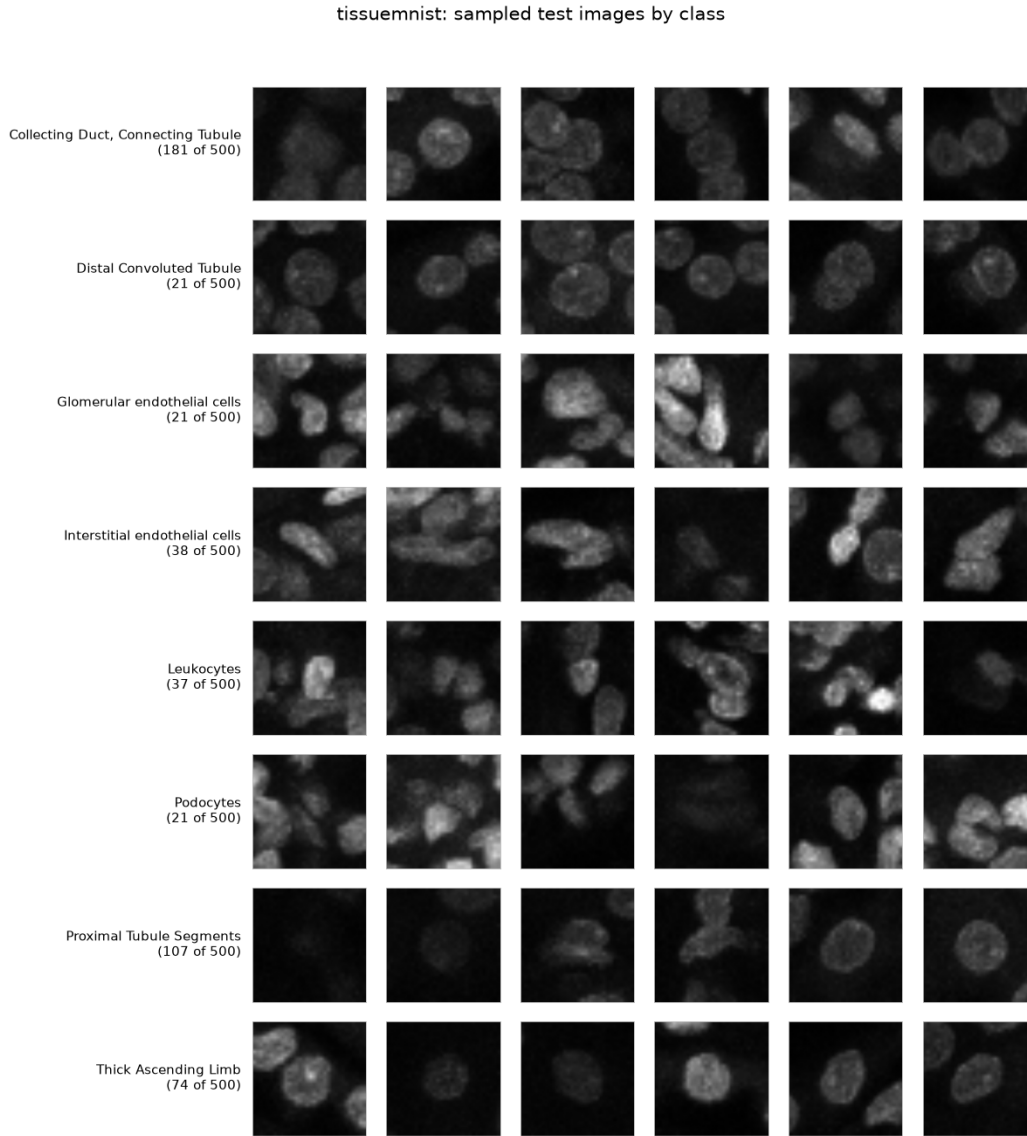


Figure 17: tissuemnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.10 organcmnist

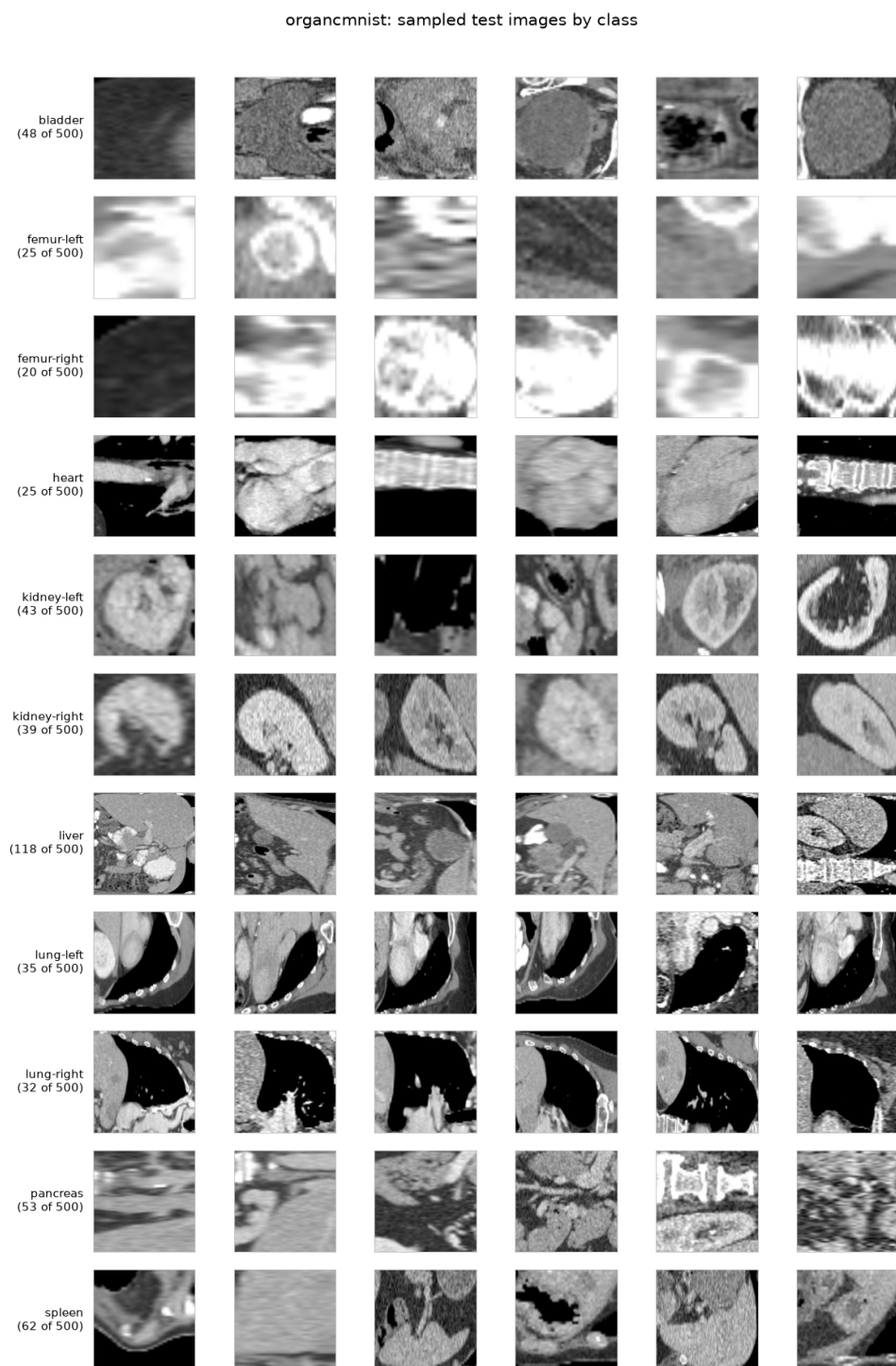


Figure 18: organcmnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.11 organismist

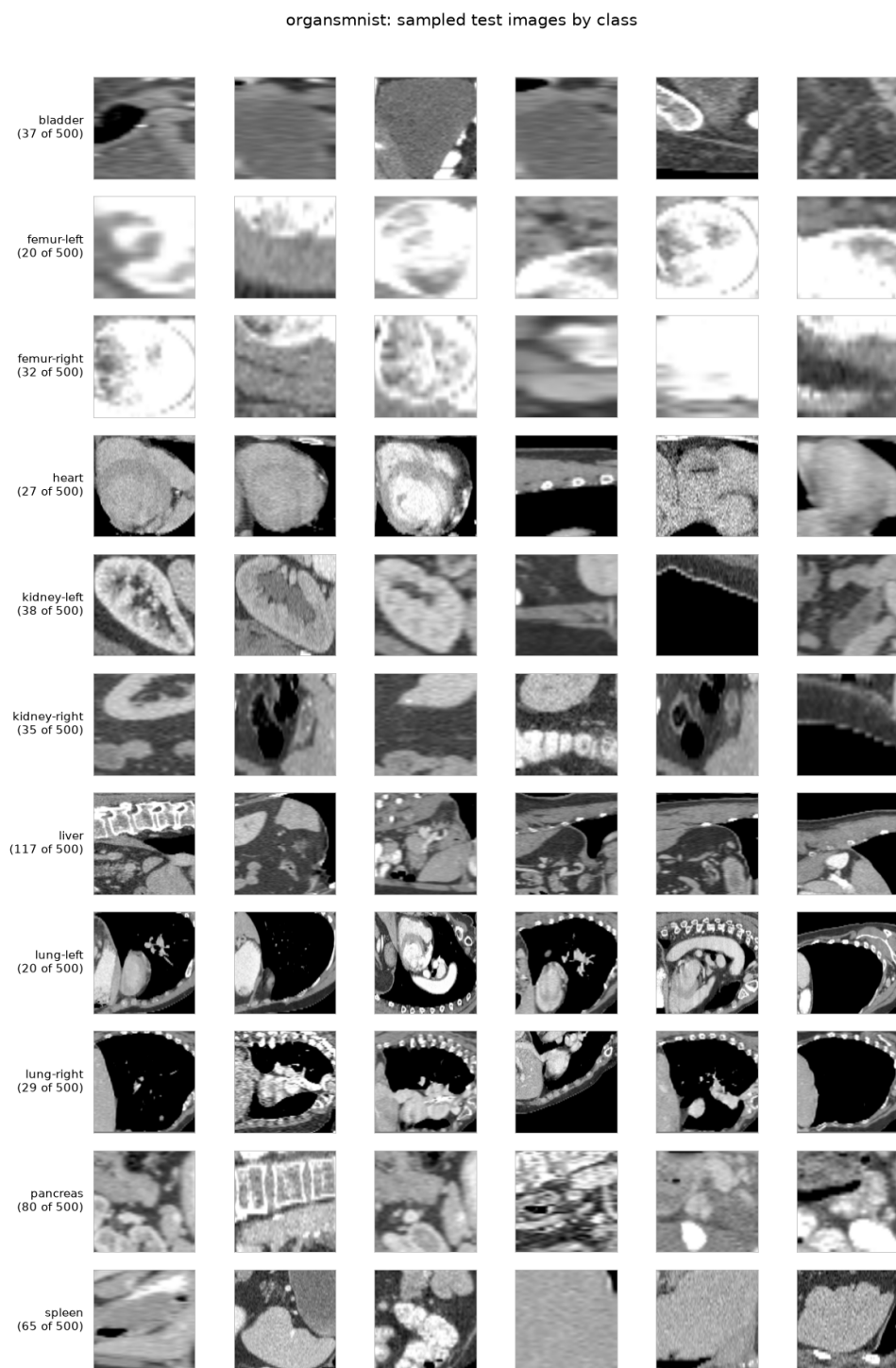


Figure 19: organsmnist: up to 6 sampled test images per class, with the count of that class in the sample.

B.12 chestmnist

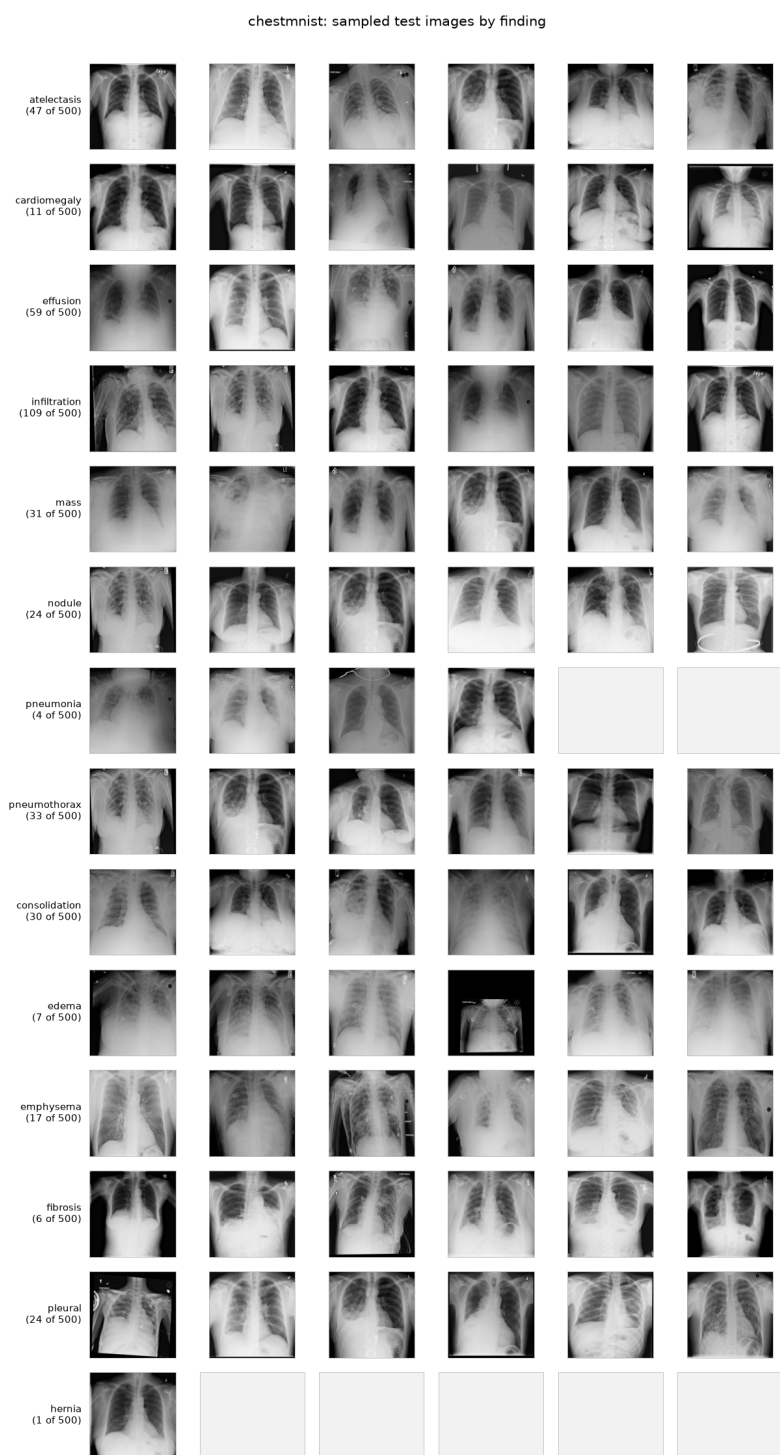


Figure 20: chestmnist: up to 6 sampled test images per finding, with the count of that finding in the sample.

C Sources behind the concept bank

References

- [1] Naheed R. Abbasi, Helen M. Shaw, Darrell S. Rigel, Robert J. Friedman, William H. McCarthy, Iman Osman, Alfred W. Kopf, and David Polsky. Early diagnosis of cutaneous melanoma: revisiting the ABCD criteria. *JAMA*, 292(22):2771–2776, 2004.
- [2] Andrea Acevedo, Santiago Alf3rez, Anna Merino, Laura Puigv3, and Jos3 Rodellar. Recognition of peripheral blood cell images using convolutional neural networks. *Computer Methods and Programs in Biomedicine*, 180:105020, 2019.
- [3] Andrea Acevedo, Anna Merino, Santiago Alf3rez, 3ngel Molina, Laura Bold3, and Jos3 Rodellar. A dataset of microscopic peripheral blood cell images for development of automatic recognition systems. *Data in Brief*, 30:105474, 2020.
- [4] Walid Al-Dhabyani, Mohammed Gomaa, Hussien Khaled, and Aly Fahmy. Dataset of breast ultrasound images. *Data in Brief*, 28:104863, 2020.
- [5] Davide Altamura, Scott W. Menzies, Giuseppe Argenziano, Iris Zalaudek, H. Peter Soyer, Francesco Sera, Michelle Avramidis, Kathryn DeAmbrosis, Maria Concetta Fargnoli, and Ketty Peris. Dermatoscopy of basal cell carcinoma: morphologic variability of global and local features and accuracy of diagnosis. *Journal of the American Academy of Dermatology*, 62(1):67–75, 2010.
- [6] Savvas Andronikou, Elena Lambert, Jarred Halton, Lucy Hilder, Iona Crumley, Mark D. Lyttle, and Cara Kosack. Guidelines for the use of chest radiographs in community-acquired pneumonia in children and adolescents. *Pediatric Radiology*, 47(11):1405–1411, 2017.
- [7] Giuseppe Argenziano, Gabriella Fabbrocini, Paolo Carli, Vincenzo De Giorgi, Elena Sammarco, and Mario Delfino. Epiluminescence microscopy for the diagnosis of doubtful melanocytic skin lesions. comparison of the ABCD rule of dermatoscopy and a new 7-point checklist based on pattern analysis. *Archives of Dermatology*, 134(12):1563–1570, 1998.
- [8] Giuseppe Argenziano, H. Peter Soyer, Sergio Chimenti, Renato Talamini, Rosamaria Corona, Francesco Sera, Michael Binder, Lorenzo Cerroni, Gaetano De Rosa, Gerardo Ferrara, Rainer Hofmann-Wellenhof, Michael Landthaler, et al. Dermoscopy of pigmented skin lesions: results of a consensus meeting via the Internet. *Journal of the American Academy of Dermatology*, 48(5):679–693, 2003.
- [9] M. Arman, M. M. A. Rahat, M. A. Thithi, J. U. Khan, S. M. Kabir, M. I. Islam, M. Hasan, M. N. Shakib, and H. S. Lim. A lightweight deep learning model with channel attention for kidney cell classification from microscopy images. *Scientific Reports*, 16:24754, 2026.
- [10] Kyongtae T. Bae. Intravenous contrast medium administration and scan timing at CT: considerations and approaches. *Radiology*, 256(1):32–61, 2010.
- [11] Alexander A. Bankier, Heber MacMahon, Thomas Colby, Pierre Alain Gevenois, Jin Mo Goo, Ann N. C. Leung, David A. Lynch, Cornelia M. Schaefer-Prokop, Noriyuki Tomiyama, William D. Travis, Johny A. Verschakelen, Charles S. White, and David P. Naidich. Fleischner Society: Glossary of terms for thoracic imaging. *Radiology*, 310(2):e232558, 2024.

- [12] Patrick Bilic, Patrick Christ, Hongwei Bran Li, Eugene Vorontsov, Avi Ben-Cohen, Georgios Kaissis, Adi Szeskin, Colin Jacobs, et al. The liver tumor segmentation benchmark (LiTS). *Medical Image Analysis*, 84:102680, 2023.
- [13] Matthias Bolz, Ursula Schmidt-Erfurth, Gabor Deak, Georgios Mylonas, Katharina Kriebbaum, and Christoph Scholda. Optical coherence tomographic hyperreflective foci: A morphologic sign of lipid extravasation in diabetic macular edema. *Ophthalmology*, 116(5):914–920, 2009.
- [14] Peyman Borghei, Farnoosh Sokhandon, Ali Shirkhoda, and Desiree E. Morgan. Anomalies, anatomic variants, and sources of diagnostic pitfalls in pancreatic imaging. *Radiology*, 266(1):28–36, 2013.
- [15] John S. Bradley, Carrie L. Byington, Samir S. Shah, Brian Alverson, Edward R. Carter, Christopher Harrison, Sheldon L. Kaplan, Sharon E. Mace, George H. McCracken, Jr., Matthew R. Moore, Shawn D. St Peter, Jana A. Stockwell, and Jack T. Swanson. The management of community-acquired pneumonia in infants and children older than 3 months of age: clinical practice guidelines by the Pediatric Infectious Diseases Society and the Infectious Diseases Society of America. *Clinical Infectious Diseases*, 53(7):e25–e76, 2011.
- [16] Ralph P. Braun, Harold S. Rabinovitz, Jean Krischer, Jürgen Kreusch, Margaret Oliviero, Luigi Naldi, Alfred W. Kopf, and Jean-Hilaire Saurat. Dermoscopy of pigmented seborrheic keratosis: a morphological study. *Archives of Dermatology*, 138(12):1556–1560, 2002.
- [17] T. Clark Brelje and Robert L. Sorenson. Urinary system: Nephrons and renal corpuscle. HistologyGuide, Department of Genetics, Cell Biology and Development, University of Minnesota, 2026. Copyright 2005–2026.
- [18] L. Bugatti and G. Filosa. Dermoscopy of lichen planus-like keratosis: a model of inflammatory regression. *Journal of the European Academy of Dermatology and Venereology*, 21(10):1392–1397, 2007.
- [19] Joseph Chapman and Yaoping Zhang. Histology, hematopoiesis. In: StatPearls. Treasure Island (FL): StatPearls Publishing, 2023. Bookshelf ID NBK534246; updated 1 May 2023.
- [20] Thomas Cherian, E. Kim Mulholland, John B. Carlin, Harald Ostensen, Ruhul Amin, Margaret de Campo, David Greenberg, Rosanna Lagos, Marilla Lucero, Shabir A. Madhi, Katherine L. O’Brien, Steven Obaro, and Mark C. Steinhoff. Standardized interpretation of paediatric chest radiographs for the diagnosis of pneumonia in epidemiological studies. *Bulletin of the World Health Organization*, 83(5):353–359, 2005.
- [21] Benie T. Constantino and Bessie Cogionis. Nucleated RBCs — significance in the peripheral blood film. *Laboratory Medicine*, 31(4):223–229, 2000.
- [22] Tami D. DenOtter and Johanna Schubert. Hounsfield unit. In *StatPearls*. StatPearls Publishing, Treasure Island, FL, 2023. Updated 2023 Mar 6.
- [23] Carl J. D’Orsi, Edward A. Sickles, Ellen B. Mendelson, and Elizabeth A. Morris. *ACR BI-RADS Atlas, Breast Imaging Reporting and Data System*. American College of Radiology, Reston, VA, 5 edition, 2013.
- [24] Early Treatment Diabetic Retinopathy Study Research Group. Grading diabetic retinopathy from stereoscopic color fundus photographs — an extension of the modified Airlie House classification. ETDRS report number 10. *Ophthalmology*, 98(5):786–806, 1991.

- [25] George W. Elgart. Seborrheic keratoses, solar lentigines, and lichenoid keratoses. dermatoscopic features and correlation to histology and clinical signs. *Dermatologic Clinics*, 19(2):347–357, 2001.
- [26] Valerie E. Espinoza and Prabhu D. Emmady. Histology, monocytes. In: StatPearls. Treasure Island (FL): StatPearls Publishing, 2023. Bookshelf ID NBK557618; updated 24 April 2023.
- [27] Nicholas Fancourt, Maria Deloria Knoll, Breanna Barger-Kamate, John de Campo, Margaret de Campo, Mahamadou Diallo, Bernard E. Ebruke, Daniel R. Feikin, Fergus Gleeson, Wenfeng Gong, Laura L. Hammitt, Rasa Izadnegahdar, Anchalee Kruatrachue, Shabir A. Madhi, Veronica Manduku, Fariha Bushra Matin, Nasreen Mahomed, David P. Moore, Musaku Mwenechanya, Kamrun Nahar, Claire Oluwalana, Micah Silaba Ominde, Christine Prosperi, Joyce Sande, Piyarat Suntarattiwong, and Katherine L. O’Brien. Standardized interpretation of chest radiographs in cases of pediatric pneumonia from the PERCH study. *Clinical Infectious Diseases*, 64(suppl_3):S253–S261, 2017.
- [28] Frederick L. Ferris, C. P. Wilkinson, Alan Bird, Usha Chakravarthy, Emily Chew, Karl Csaky, and Srinivas R. Sadda. Clinical classification of age-related macular degeneration. *Ophthalmology*, 120(4):844–851, 2013.
- [29] J. T. Flaherty, J. E. Pierce, V. J. Ferrans, D. J. Patel, W. K. Tucker, and D. L. Fry. Endothelial nuclear patterns in the canine arterial tree with particular reference to hemodynamic events. *Circulation Research*, 30(1):23–33, 1972.
- [30] Christina J. Flaxel, Ron A. Adelman, Steven T. Bailey, Amani Fawzi, Jennifer I. Lim, G. Atma Vemulakonda, and Gui-shuang Ying. Diabetic retinopathy preferred practice pattern. *Ophthalmology*, 127(1):P66–P145, 2020.
- [31] Matthew Fleming, Sreelakshmi Ravula, Sergei F. Tatishchev, and Hanlin L. Wang. Colorectal carcinoma: Pathologic aspects. *Journal of Gastrointestinal Oncology*, 3(3):153–173, 2012.
- [32] Thomas Gluecker, Patrizio Capasso, Pierre Schnyder, François Gudinchet, Marie-Denise Schaller, Jean-Pierre Revelly, René Chiolerio, Peter Vock, and Stephan Wicky. Clinical and radiologic features of pulmonary edema. *RadioGraphics*, 19(6):1507–1531, 1999.
- [33] David M. Hansell, Alexander A. Bankier, Heber MacMahon, Theresa C. McLoud, Nestor L. Müller, and Jacques Remy. Fleischner Society: Glossary of terms for thoracic imaging. *Radiology*, 246(3):697–722, 2008.
- [34] Maged Hassan, Abdelfattah A. Touman, Elżbieta M. Grabczak, Søren H. Skaarup, Katarzyna Faber, Kevin G. Blyth, and Svitlana Pochepnia. Imaging of pleural disease. *Breathe*, 20(1):230172, 2024.
- [35] Andrea S. Hong, Eric L. Rosen, Mary S. Soo, and Jay A. Baker. BI-RADS for sonography: positive and negative predictive values of sonographic features. *American Journal of Roentgenology*, 184(4):1260–1265, 2005.
- [36] Brigitte Kaissling and Michel Le Hir. The renal cortical interstitium: Morphological and functional aspects. *Histochemistry and Cell Biology*, 130(2):247–262, 2008.
- [37] Jakob Nikolas Kather, Johannes Krisam, Pornpimol Charoentong, Tom Luedde, Esther Herpel, Cleo-Aron Weis, Timo Gaiser, Alexander Marx, Nektarios A. Valous, Dyke Ferber, Lina Jansen, Constantino Carlos Reyes-Aldasoro, Inka Zörnig, Dirk Jäger, Hermann Brenner, Jenny

- Chang-Claude, Michael Hoffmeister, and Niels Halama. Predicting survival from colorectal cancer histology slides using deep learning: A retrospective multicenter study. *PLOS Medicine*, 16(1):e1002730, 2019.
- [38] Jakob Nikolas Kather, Cleo-Aron Weis, Francesco Bianconi, Susanne M. Melchers, Lothar R. Schad, Timo Gaiser, Alexander Marx, and Frank Gerrit Zöllner. Multi-class texture analysis in colorectal cancer histology. *Scientific Reports*, 6:27988, 2016.
- [39] Yusuke Kawasaki, Yoshihiro Hosoyamada, Takayuki Miyaki, Junji Yamaguchi, Soichiro Kakuta, Tatsuo Sakai, and Koichiro Ichimura. Three-dimensional architecture of glomerular endothelial cells revealed by FIB-SEM tomography. *Frontiers in Cell and Developmental Biology*, 9:653472, 2021.
- [40] Pearse A. Keane, Praveen J. Patel, Sandra Liakopoulos, Florian M. Heussen, Srinivas R. Sadda, and Adnan Tufail. Evaluation of age-related macular degeneration with optical coherence tomography. *Survey of Ophthalmology*, 57(5):389–414, 2012.
- [41] Daniel S. Kermany, Michael Goldbaum, Wenjia Cai, Carolina C. S. Valentim, Huiying Liang, Sally L. Baxter, Alex McKeown, Ge Yang, Xiaokang Wu, Fangbing Yan, Justin Dong, Made K. Prasadha, Jacqueline Pei, Magdalene Y. L. Ting, Jie Zhu, Christina Li, Sierra Hewett, Jason Dong, Ian Ziyar, Alexander Shi, Runze Zhang, Lianghong Zheng, Rui Hou, William Shi, Xin Fu, Yaou Duan, Viet A. N. Huu, Cindy Wen, Edward D. Zhang, Charlotte L. Zhang, Oulan Li, Xiaobo Wang, Michael A. Singer, Xiaodong Sun, Jie Xu, Ali Tafreshi, M. Anthony Lewis, Huimin Xia, and Kang Zhang. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5):1122–1131.e9, 2018.
- [42] Aziz A. Khanifar, Anjum F. Koreishi, Joseph A. Izatt, and Cynthia A. Toth. Drusen ultrastructure imaging with spectral domain optical coherence tomography in age-related macular degeneration. *Ophthalmology*, 115(11):1883–1890.e1, 2008.
- [43] David W. King. Kidney: Epithelium and connective tissue; renal cortex: Proximal and distal tubules. Histology at SIU, Southern Illinois University School of Medicine, 2022. Pages RN006c (updated 8 June 2022) and RN008b (updated 30 May 2022).
- [44] David W. King. Renal system: Guide to the identification of renal tubules. Histology at SIU, Southern Illinois University School of Medicine, 2025. Updated 12 November 2025.
- [45] Lucy Kirkwood. Kidney: Normal renal histology. Pathologia, University of Edinburgh, 2017. Undated page; image credits dated 2017.
- [46] Harald Kittler, Ashfaq A. Marghoob, Giuseppe Argenziano, Cristina Carrera, Clara Curiel-Lewandrowski, Rainer Hofmann-Wellenhof, Josep Malvehy, Scott Menzies, Susana Puig, Harold Rabinovitz, Wilhelm Stolz, Toshiaki Saida, et al. Standardization of terminology in dermoscopy/dermatoscopy: results of the third consensus conference of the International Society of Dermoscopy. *Journal of the American Academy of Dermatology*, 74(6):1093–1106, 2016.
- [47] Kai Klintrup, Johanna M. Mäkinen, Saira Kauppila, Päivi O. Väre, Jukka Melkko, Hannu Tuominen, Karoliina Tuppurainen, Jyrki Mäkelä, Tuomo J. Karttunen, and Markus J. Mäkinen. Inflammation and prognosis in colorectal cancer. *European Journal of Cancer*, 41(17):2645–2654, 2005.
- [48] Jiyon Lee. Practical and illustrated summary of updated BI-RADS for ultrasonography. *Ultrasonography*, 36(1):71–81, 2017.

- [49] Ruhan Liu, Xiangning Wang, Qiang Wu, Ling Dai, Xi Fang, Tao Yan, Jaemin Son, Shiqi Tang, Jiang Li, Zijian Gao, Adrian Galdran, J. M. Poorneshwaran, Hao Liu, Jie Wang, Yerui Chen, Prasanna Porwal, Gavin Siew Wei Tan, Xiaokang Yang, Chao Dai, Haitao Song, Mingang Chen, Huating Li, Weiping Jia, Dinggang Shen, Bin Sheng, and Ping Zhang. DeepDRiD: Diabetic retinopathy — grading and image quality estimation challenge. *Patterns*, 3(6):100512, 2022.
- [50] Vebjorn Ljosa, Katherine L. Sokolnicki, and Anne E. Carpenter. Annotated high-throughput microscopy image sets for validation. *Nature Methods*, 9(7):637, 2012.
- [51] Antonio Madrazo-Ibarra and Pradeep Vaitla. Histology, nephron. StatPearls [Internet], StatPearls Publishing, Treasure Island (FL). Bookshelf ID NBK554411, 2023. Updated 17 February 2023.
- [52] Harriet R. Manley, Maria C. Keightley, and Graham J. Lieschke. The neutrophil nucleus: An important influence on neutrophil migration and function. *Frontiers in Immunology*, 9:2867, 2018.
- [53] Ellen B. Mendelson, Marcela Böhm-Vélez, Wendie A. Berg, et al. ACR BI-RADS ultrasound. In *ACR BI-RADS Atlas, Breast Imaging Reporting and Data System*. American College of Radiology, Reston, VA, 5 edition, 2013.
- [54] Scott W. Menzies, Karin Westerhoff, Harold Rabinovitz, Alfred W. Kopf, William H. McCarthy, and Brian Katz. Surface microscopy of pigmented basal cell carcinoma. *Archives of Dermatology*, 136(8):1012–1016, 2000.
- [55] Anthony L. Mescher. *Junqueira’s Basic Histology: Text and Atlas*. McGraw Hill, New York, 16 edition, 2021.
- [56] Anna Rita Migliaccio. Erythroblast enucleation. *Haematologica*, 95(12):1985–1988, 2010.
- [57] Torsten B. Moeller and Emil Reif. *Pocket Atlas of Sectional Anatomy: Computed Tomography and Magnetic Resonance Imaging. Volume II: Thorax, Heart, Abdomen and Pelvis*. Thieme, Stuttgart, 4 edition, 2013.
- [58] Ian V. Murray and Michael A. Paolini. Histology, kidney and glomerulus. StatPearls [Internet], StatPearls Publishing, Treasure Island (FL). Bookshelf ID NBK554544, 2023. Updated 17 April 2023.
- [59] Franz Nachbar, Wilhelm Stolz, Tanja Merkle, Armand B. Cognetta, Thomas Vogt, Michael Landthaler, Peter Bilek, Otto Braun-Falco, and Gerd Plewig. The ABCD rule of dermatoscopy. high prospective value in the diagnosis of doubtful melanocytic skin lesions. *Journal of the American Academy of Dermatology*, 30(4):551–559, 1994.
- [60] Laura K. Nason, Christopher M. Walker, Michael F. McNeeley, Wanaporn Burivong, Corinne L. Fligner, and J. David Godwin. Imaging of the diaphragm: Anatomy and function. *RadioGraphics*, 32(2):E51–E70, 2012.
- [61] Tomohiro Otani, Shoji Kishi, and Yasuhiro Maruyama. Patterns of diabetic macular edema with optical coherence tomography. *American Journal of Ophthalmology*, 127(6):688–693, 1999.
- [62] L. Palmer, C. Briggs, S. McFadden, G. Zini, J. Burthem, G. Rozenberg, M. Proytcheva, and S. J. Machin. ICSH recommendations for the standardization of nomenclature and grading of peripheral blood cell morphological features. *International Journal of Laboratory Hematology*, 37(3):287–303, 2015.

- [63] Hermann Pavenstädt, Wilhelm Kriz, and Matthias Kretzler. Cell biology of the glomerular podocyte. *Physiological Reviews*, 83(1):253–307, 2003.
- [64] Marion J. Pollheimer, Peter Kornprat, Richard A. Lindtner, Lars Harbaum, Andrea Schlemmer, Peter Rehak, and Cord Langner. Tumor necrosis is a new promising prognostic factor in colorectal cancer. *Human Pathology*, 41(12):1749–1757, 2010.
- [65] Ajay Aroor Rao, Jennifer Feneis, Chloe Lalonde, and Haydee Ojeda-Fournier. A pictorial review of changes in the BI-RADS fifth edition. *RadioGraphics*, 36(3):623–639, 2016.
- [66] J. Rodellar, S. Alférez, A. Acevedo, A. Molina, and A. Merino. Image processing and machine learning in the morphological analysis of blood cells. *International Journal of Laboratory Hematology*, 40(S1):46–53, 2018.
- [67] Mustafa Turhan Şahin, Serap Öztürkcan, Aylin Türel Ermertcan, and Ali Tahsin Güneş. A comparison of dermoscopic features among lentigo senilis/initial seborrheic keratosis, seborrheic keratosis, lentigo maligna and lentigo maligna melanoma on the face. *The Journal of Dermatology*, 31(11):884–889, 2004.
- [68] Sheila Sheth and Elliot K. Fishman. Multi-detector row CT of the kidneys and urinary tract: techniques and applications in the diagnosis of benign diseases. *RadioGraphics*, 24(2):e20, 2004.
- [69] Richard F. Spaide, Glenn J. Jaffe, David Sarraf, K. Bailey Freund, Srinivas R. Sadda, Giovanni Staurenghi, Nadia K. Waheed, Usha Chakravarthy, Philip J. Rosenfeld, Frank G. Holz, Eric H. Souied, Salomon Y. Cohen, Giuseppe Querques, Kyoko Ohno-Matsui, David Boyer, Alain Gaudric, Barbara Blodi, Caroline R. Bauman, Xiaoxin Li, Gabriel J. Coscas, Alexander Brucker, Lawrence Singerman, Philip Luthert, Steffen Schmitz-Valckenberg, Ursula Schmidt-Erfurth, Hans E. Grossniklaus, David J. Wilson, Robyn Guymer, Lawrence A. Yannuzzi, Emily Y. Chew, Karl Csaky, Jordi M. Mones, Daniel Pauleikhoff, Ramin Tadayoni, and James Fujimoto. Consensus nomenclature for reporting neovascular age-related macular degeneration data: Consensus on neovascular age-related macular degeneration nomenclature study group. *Ophthalmology*, 127(5):616–636, 2020.
- [70] Jonathan D. Spratt, Lonie R. Salkowski, Marios Loukas, Tom Turmezei, Jamie Weir, and Peter H. Abrahams. *Weir & Abrahams’ Imaging Atlas of Human Anatomy*. Elsevier, Edinburgh, 5 edition, 2016.
- [71] Giovanni Staurenghi, Srinivas Sadda, Usha Chakravarthy, and Richard F. Spaide. Proposed lexicon for anatomic landmarks in normal posterior segment spectral-domain optical coherence tomography: The IN•OCT consensus. *Ophthalmology*, 121(8):1572–1578, 2014.
- [72] A. Thomas Stavros, David Thickman, Cynthia L. Rapp, Mark A. Dennis, Steve H. Parker, and Gale A. Sisney. Solid breast nodules: use of sonography to distinguish between benign and malignant lesions. *Radiology*, 196(1):123–134, 1995.
- [73] William M. Thurlbeck and George Simon. Radiographic appearance of the chest in emphysema. *American Journal of Roentgenology*, 130(3):429–440, 1978.
- [74] Alyssa Tigner, Sherif A. Ibrahim, and Ian V. Murray. Histology, white blood cell. In: StatPearls. Treasure Island (FL): StatPearls Publishing, 2026. Bookshelf ID NBK563148; updated 14 February 2026.
- [75] Krystian Truskiewicz, Rafał Poręba, and Paweł Gać. Radiological cardiothoracic ratio in evidence-based medicine. *Journal of Clinical Medicine*, 10(9):2016, 2021.

- [76] Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5(1):180161, 2018.
- [77] Hideki Ueno, Yoshiki Kajiwar, Yoichi Ajioka, Tamotsu Sugai, Shigeki Sekine, Megumi Ishiguro, Atsuo Takashima, and Yukihide Kanemitsu. Histopathological atlas of desmoplastic reaction characterization in colorectal cancer. *Japanese Journal of Clinical Oncology*, 51(6):1004–1012, 2021.
- [78] Vighnesh Walavalkar. Kidney biopsy of the month — the tubulointerstitium part 1: The cortex. Renal Fellow Network, 2019. Posted 11 February 2019.
- [79] Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. ChestX-ray8: Hospital-scale chest X-Ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3462–3471. IEEE, 2017.
- [80] C. P. Wilkinson, Frederick L. Ferris, III, Ronald E. Klein, Paul P. Lee, Carl David Agardh, Matthew Davis, Diana Dills, Anselm Kampik, R. Pararajasegaram, and Juan T. Verdaguer. Proposed international clinical diabetic retinopathy and diabetic macular edema disease severity scales. *Ophthalmology*, 110(9):1677–1682, 2003.
- [81] Olivia Williams and Shane R. Sergent. Histology, platelets. In: StatPearls. Treasure Island (FL): StatPearls Publishing, 2022. Bookshelf ID NBK557800; updated 14 November 2022.
- [82] A. Woloshuk, S. Khochare, A. F. Almulhim, A. T. McNutt, D. Dean, D. Barwinska, M. J. Ferkowicz, M. T. Eadon, K. J. Kelly, K. W. Dunn, M. A. Hasan, T. M. El-Achkar, and S. Winfree. In situ classification of cell types in human kidney tissue using 3D nuclear staining. *Cytometry Part A*, 99(7):707–721, 2021.
- [83] Tien Y. Wong, Chui Ming Gemmy Cheung, Michael Larsen, Sanjay Sharma, and Rafael Simó. Diabetic retinopathy. *Nature Reviews Disease Primers*, 2:16012, 2016.
- [84] Tien Y. Wong, Jennifer Sun, Ryo Kawasaki, Paisan Ruamviboonsuk, Neeru Gupta, Van Charles Lansingh, Mauricio Maia, Wanjiku Mathenge, Sunil Moreker, Mahi M. K. Muqit, Serge Resnikoff, Juan Verdaguer, Peiquan Zhao, Frederick Ferris, Lloyd Paul Aiello, and Hugh R. Taylor. Guidelines on diabetic eye care: The International Council of Ophthalmology recommendations for screening, follow-up, referral, and treatment based on resource settings. *Ophthalmology*, 125(10):1608–1622, 2018.
- [85] John H. Woodring and James C. Reed. Radiographic manifestations of lobar atelectasis. *Journal of Thoracic Imaging*, 11(2):109–144, 1996.
- [86] World Health Organization Pneumonia Vaccine Trial Investigators’ Group. Standardization of interpretation of chest radiographs for the diagnosis of pneumonia in children. Report WHO/V&B/01.35, 2001.
- [87] Xuanang Xu, Fugen Zhou, Bo Liu, Dongshan Fu, and Xiangzhi Bai. Efficient multiple organ localization in CT image using 3D region proposal network. *IEEE Transactions on Medical Imaging*, 38(8):1885–1898, 2019.

- [88] Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. MedMNIST v2 – a large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10(1):41, 2023.
- [89] Pedro Zaballos, S. Blazquez, Susana Puig, E. Salsench, J. Rodero, J. M. Vives, and Josep Malvehy. Dermoscopic pattern of intermediate stage in seborrhoeic keratosis regressing to lichenoid keratosis: report of 24 cases. *British Journal of Dermatology*, 157(2):266–272, 2007.
- [90] Pedro Zaballos, Cinta Daufí, Susana Puig, Giuseppe Argenziano, David Moreno-Ramírez, Horacio Cabo, Ashfaq A. Marghoob, Àlex Llambrich, Iris Zalaudek, and Josep Malvehy. Dermoscopy of solitary angiokeratomas: a morphological study. *Archives of Dermatology*, 143(3):318–325, 2007.
- [91] Pedro Zaballos, Susana Puig, Alex Llambrich, and Josep Malvehy. Dermoscopy of dermatofibromas: a prospective morphological study of 412 cases. *Archives of Dermatology*, 144(1):75–83, 2008.
- [92] Iris Zalaudek, Jason Giacomel, Karin Schmid, Silvia Bondino, Cliff Rosendahl, Stefano Cavicchini, Athanasia Tournalaki, Saturnino Gasparini, Peter Bourne, Jeff Keir, Harald Kittler, Laura Eibenschutz, et al. Dermatoscopy of facial actinic keratosis, intraepidermal carcinoma, and invasive squamous cell carcinoma: a progression model. *Journal of the American Academy of Dermatology*, 66(4):589–597, 2012.